Introduction to
Finite Element Method

Xiaogai Li

Assoc. Prof. at Division of Neuronic Engineering
Department of Biomedical Engineering and Health Systems
CBH, KTH

Finite Element Method

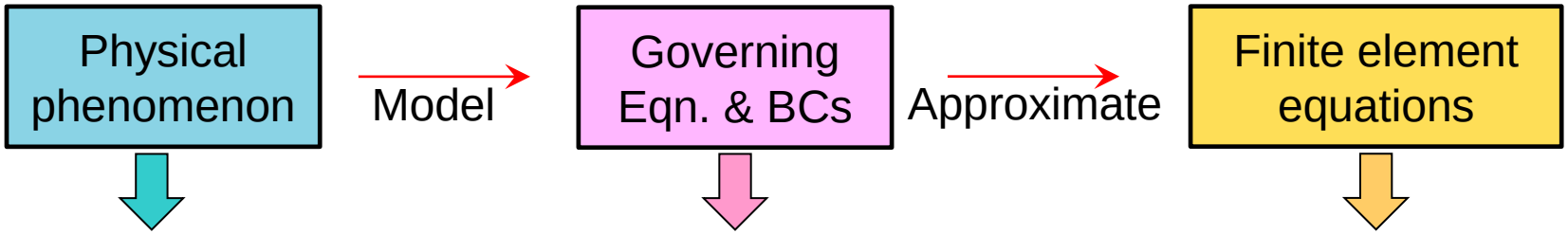
Understand:

- The fundamentals of FEM
- Applications in brain injury/prevention (what FEM can do/how)

Schedule:

- Part 1: Applications and practices
- Part 2: Perform a FEA by hand for a 1-D problem

FEM? numerical meth. to solve differential equations



- Solid Mechanics

e.g. Axially loaded elastic bar

$$\frac{d}{dx}\left(EA\frac{du}{dx}\right)+q=0$$

- Fluid Mechanics

e.g. Poiseuille flow in pipe

$$\frac{d}{dx}\left(A\frac{D^2}{32\mu}\frac{dp}{dx}\right)+Q=0$$

- Diffusion

e.g. 1-D diffusion

$$\frac{d}{dx}\left(AD\frac{dC}{dx}\right)+Q=0$$

- Thermal conduction

e.g. 1-D heat flow

$$\frac{d}{dx}\left(Ak\frac{dT}{dx}\right)+Q=0$$

- Electrical conduction

e.g. 1-D electric current flow

$$\frac{d}{dx}\left(A\sigma\frac{dV}{dx}\right)+Q=0$$

+BCs
(Boundary Conditions)

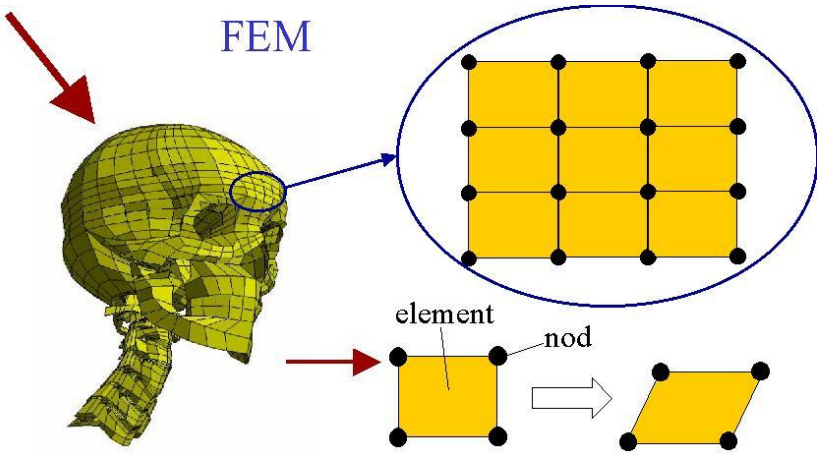
System of eqns:

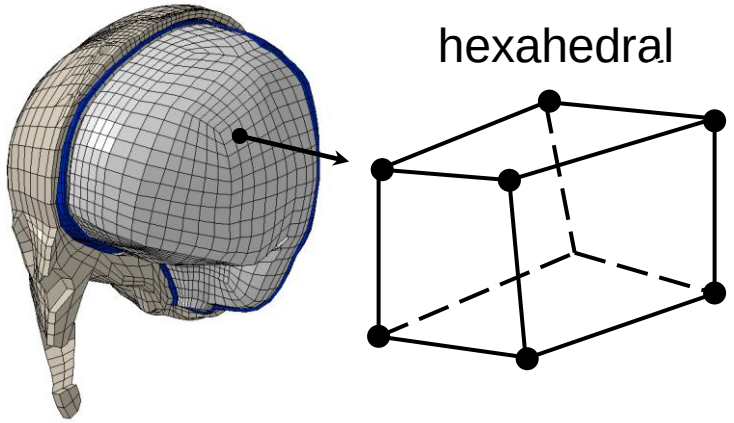
$$\mathbf{KU}=\mathbf{F}$$

Stiffness matrix | Nodal disp. | Nodal forces

Finite elements

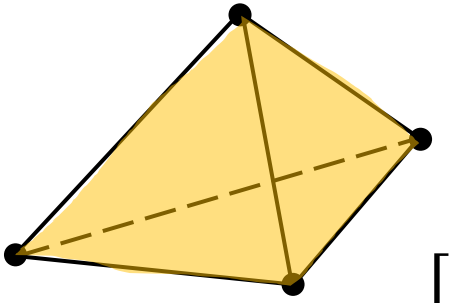
- Continous domains (geometries) -> discrete **elements** connected by **nodes**





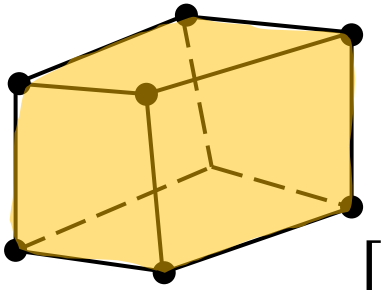
Finite elements

tetrahedral



at each node $\mathbf{U} = \begin{bmatrix} ux \\ uy \\ uz \end{bmatrix}$
(3 degree of freedom)

hexahedral



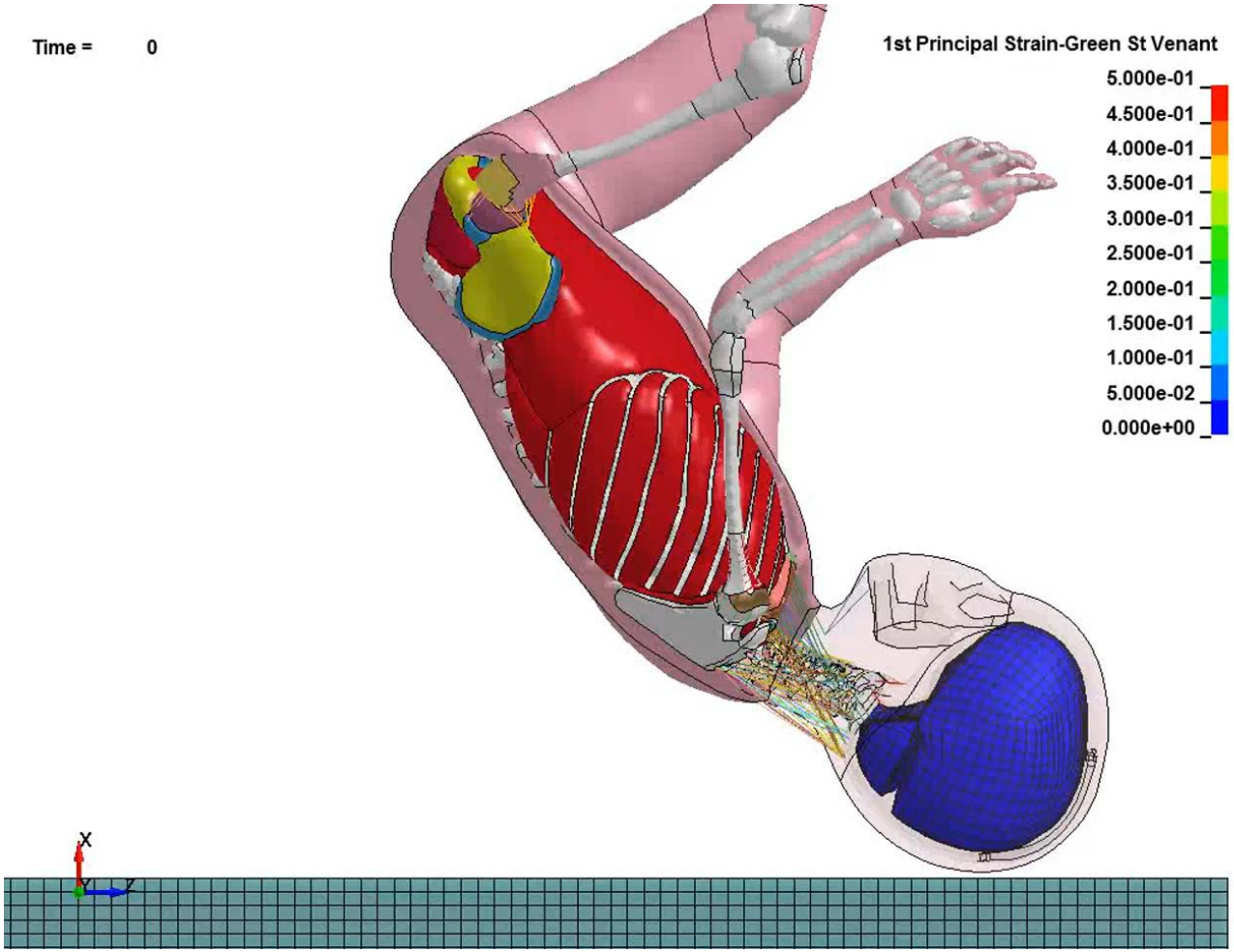
at each node $\mathbf{U} = \begin{bmatrix} ux \\ uy \\ uz \end{bmatrix}$
(3 degree of freedom)

$$\mathbf{U} = \mathbf{K}^{-1}\mathbf{F}$$

- Get solution of field varibles (disp.) on nodes
- Solution of field varibles (disp.) between nodes using interpolation/shape functions

Example videos

The video below has been produced from finite element simulations



>500,000 elements

>140000 nodes

Li X, Kleiven S. Scientific Reports. 2018 Oct 10;8(1):1-2.

Images from ’SVT’ and ’Sportnytt’ 2004-08-12


TCR. 00:00:19:03*
V
A

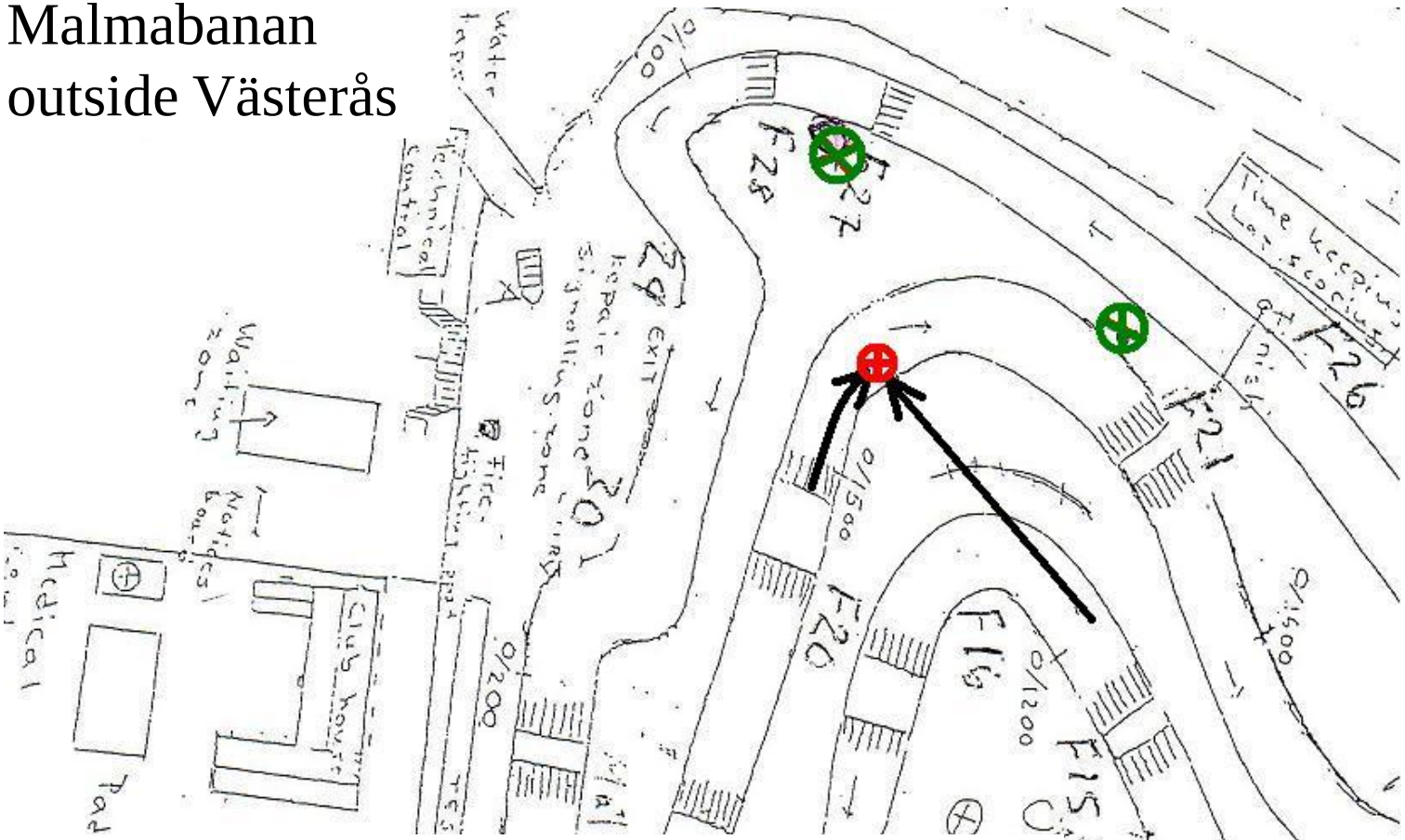
Images from the accident



40 ms between the images

The accident

Malmabanan
outside Västerås



The velocity was ca 50km/h for both riders.

The impact was almost perpendicular.

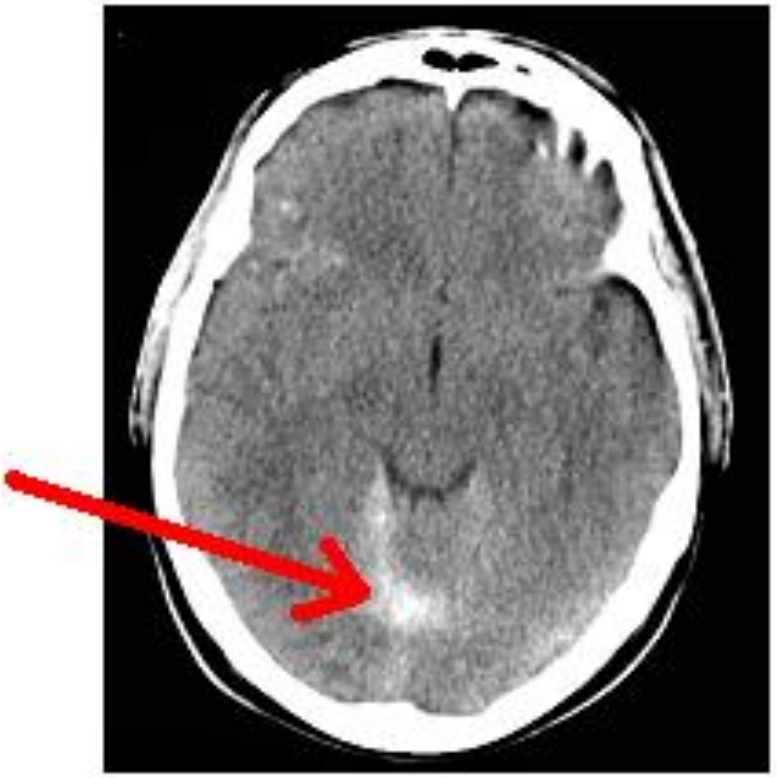
Two cameras documented the accident.

Injuries

CT images from Akademiska Sjukhuset, Uppsala



Hematoma in frontal lobe



Hematoma along the tentorium

The motorcycle



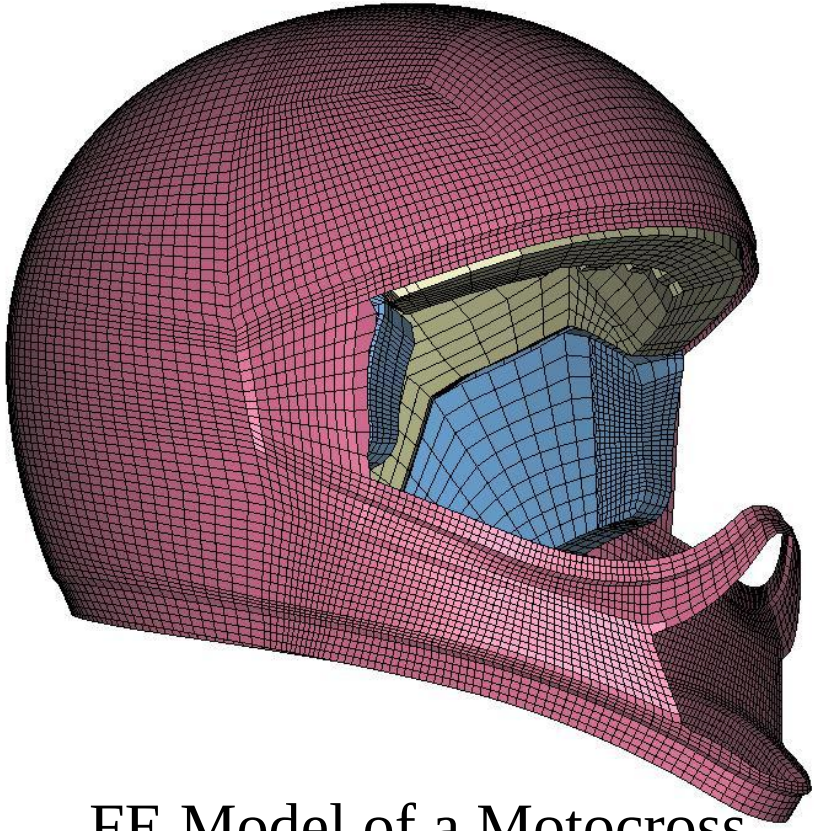
Mass of ca 110 kg



Front fender of ABS-plastics

The helmet





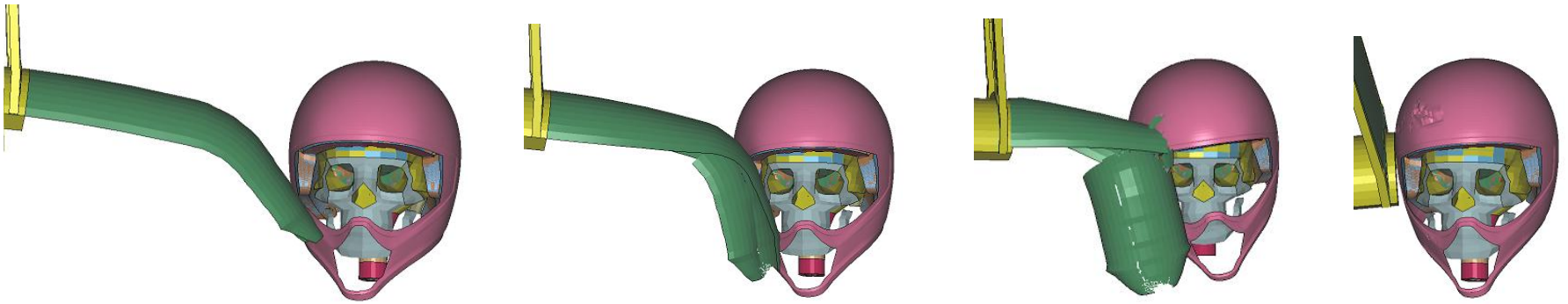
FE Model of a Motocross helmet, including Glass-fiber shell and EPS-foam

Simulations

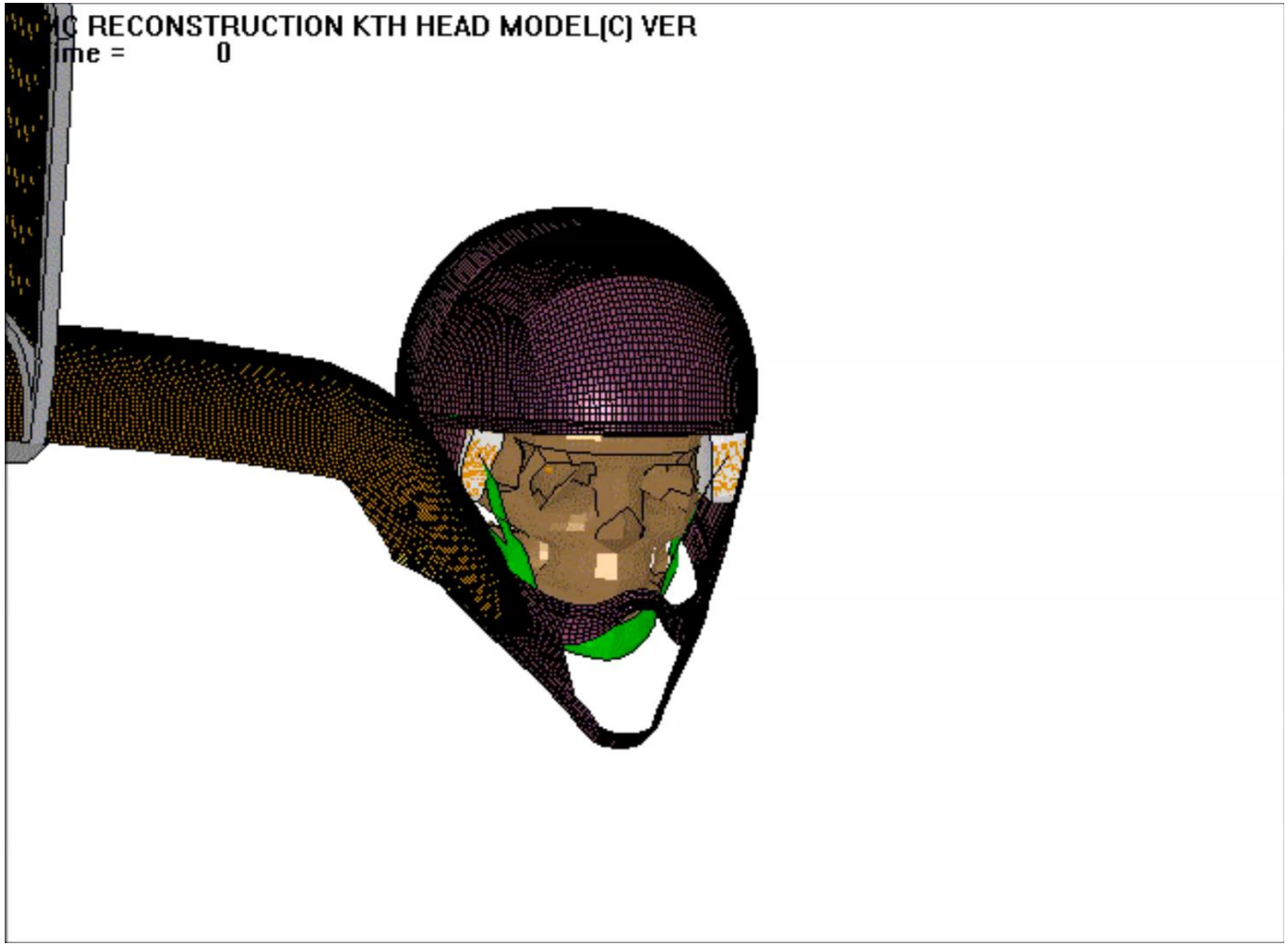
The accident was simulated using two types of helmets.

1. A conventionel helmet, where the helmet shell is glued on the foam.
2. A ”MIPS-helmet”, where the helmet shell can move relative to the rest of the helmet during impact.

   The simulations were identical, apart from using helmets of different design.



Movie from the simulations



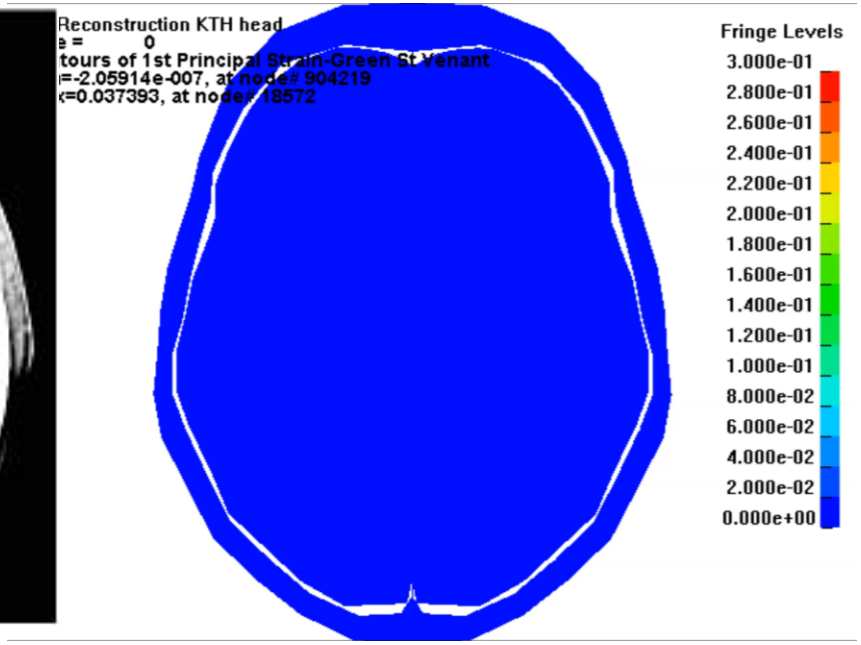
Kleiven, *World Congress Biomechanics 2006*

Strain & pressure pattern in the brain

Hematoma in
the frontal lobe

FEM-simulations





CT image

Strain

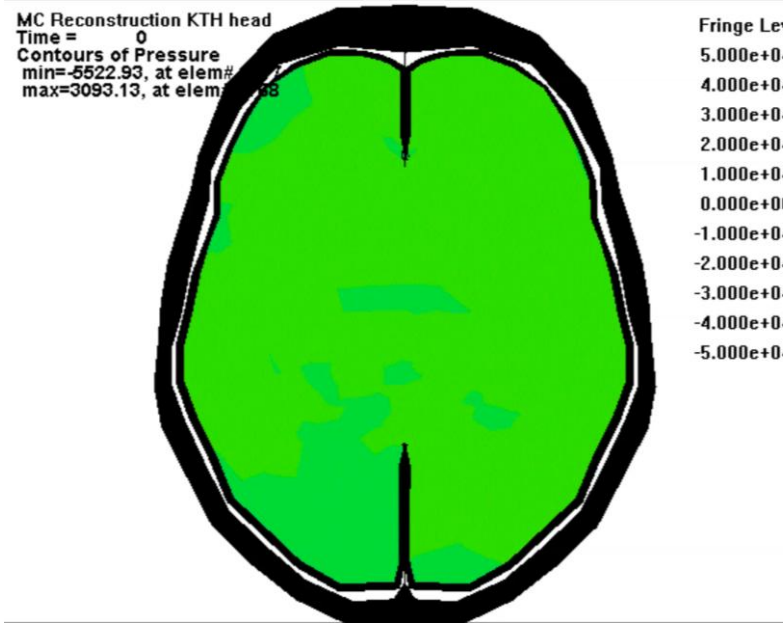
Strain & pressure pattern in the brain

Hematoma in
the frontal lobe



CT image

FEM-simulations



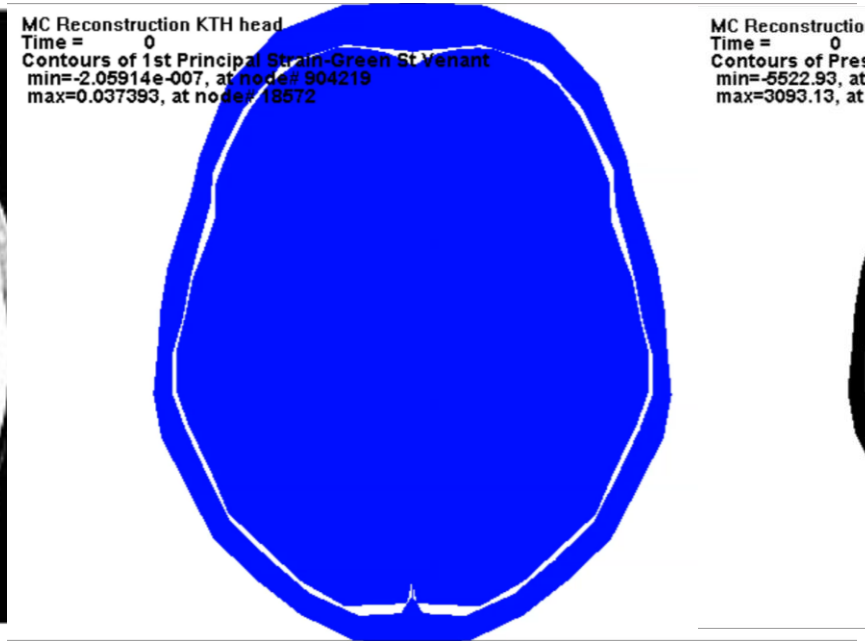
Pressure

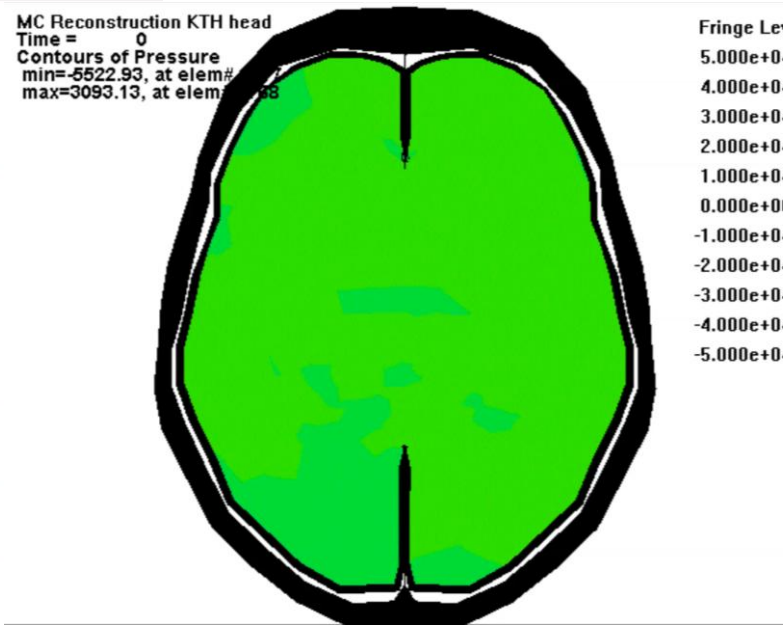
Strain & pressure pattern in the brain

Hematoma in
the frontal lobe

FEM-simulations







CT image

Strain

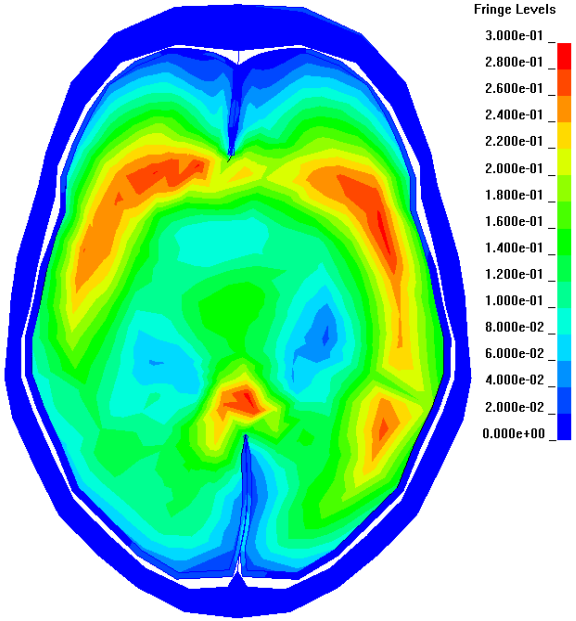
Pressure

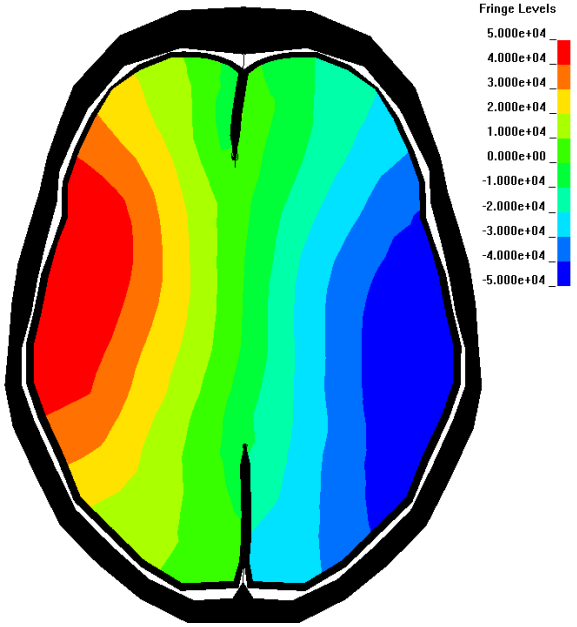
Strain & pressure pattern in the brain

Hematoma in
the frontal lobe

FEM-simulations







CT image

Strain

Pressure

Strain pattern in the brain

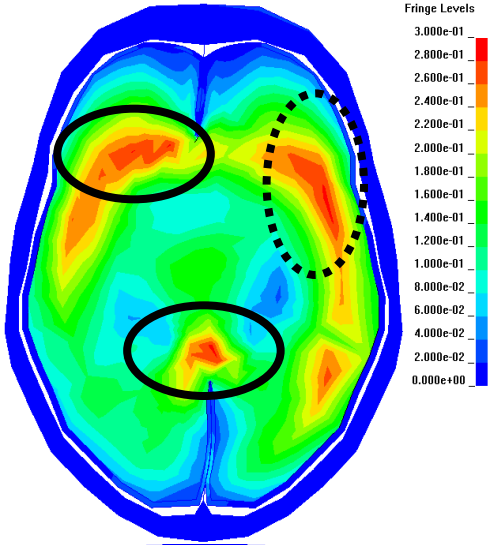
CT images

FEM-simulations

Large strains are seen in red

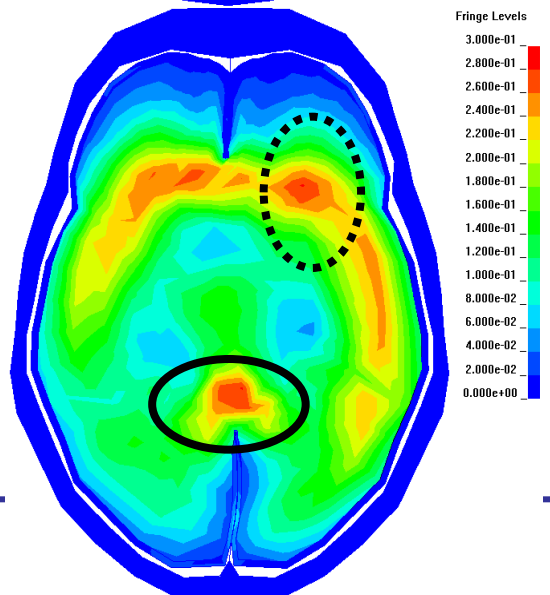
Hematoma in the frontal lobe





Hematoma in the rear part of the brain





Modifying the helmet design

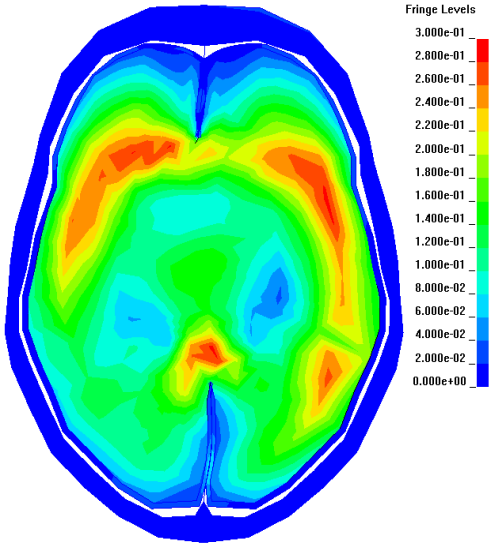
Hematoma in the frontal lobe

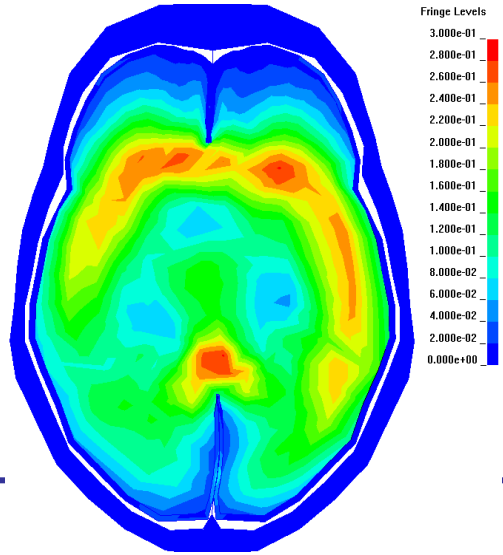


Hematoma in the rear part of the brain

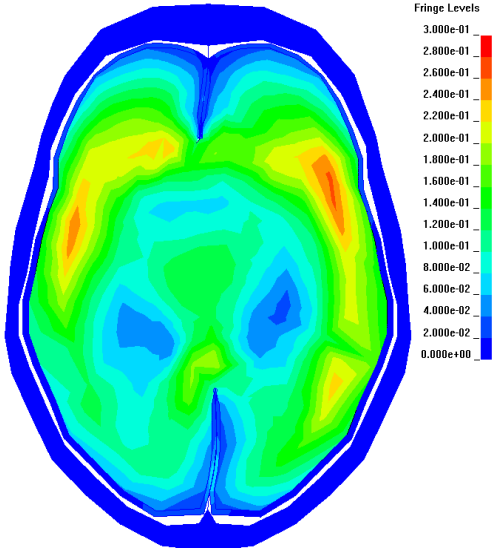


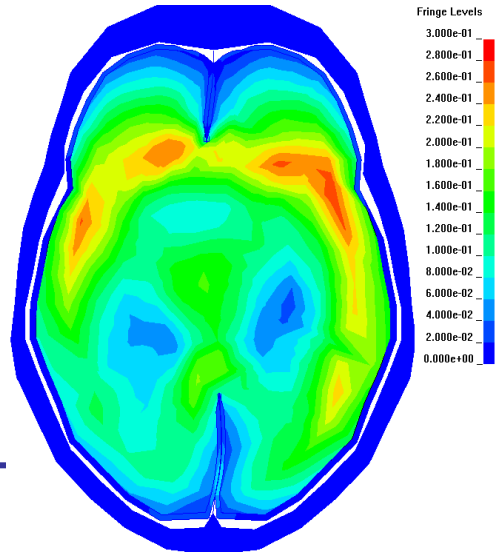
Regular helmet design





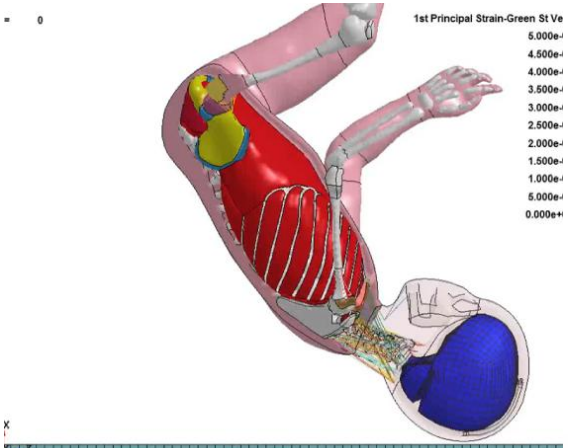
Modified helmet design

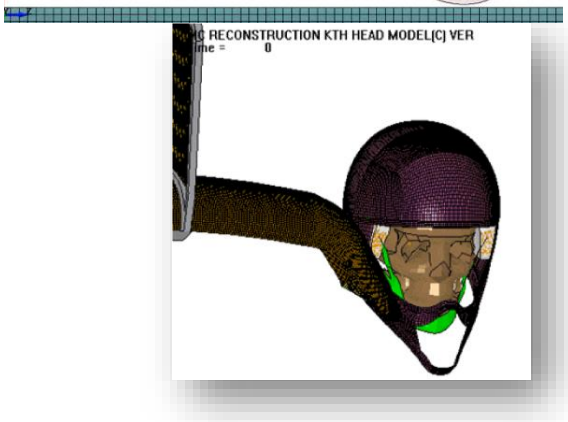






Impact





The impact can be described by a set **governing equations** - PDEs (e.g. 3D linear elastic)

Equilibrium $(F = ma)$

$$\frac{\partial \sigma_{ij}}{\partial x_i} + \rho_0 b_j = \rho_0 a_j$$

Constitutive eqns. Stress-strain

$$\sigma_{ij} = \frac{E}{1+v}\left\{\varepsilon_{ij} + \frac{v}{1-2v}\varepsilon_{kk}\delta_{ij}\right\}$$

Displacement-strain

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

Loading&BC



Impact

Displacement **U** ⟶ Strain, Stress, Pressure

The impact can be described by a set **governing equations** - PDEs (e.g. 3D linear elastic)

$$-\frac{E}{2(1+v)}\Delta \mathrm{U} - \left(\frac{E}{2(1+v)(1-2v)}\right)\nabla(\nabla\cdot \mathrm{U}) = f$$

Loading&BC

Find U=U(x,y,z) satisfies the PDEs

FEM

Approximate

System of eqns:

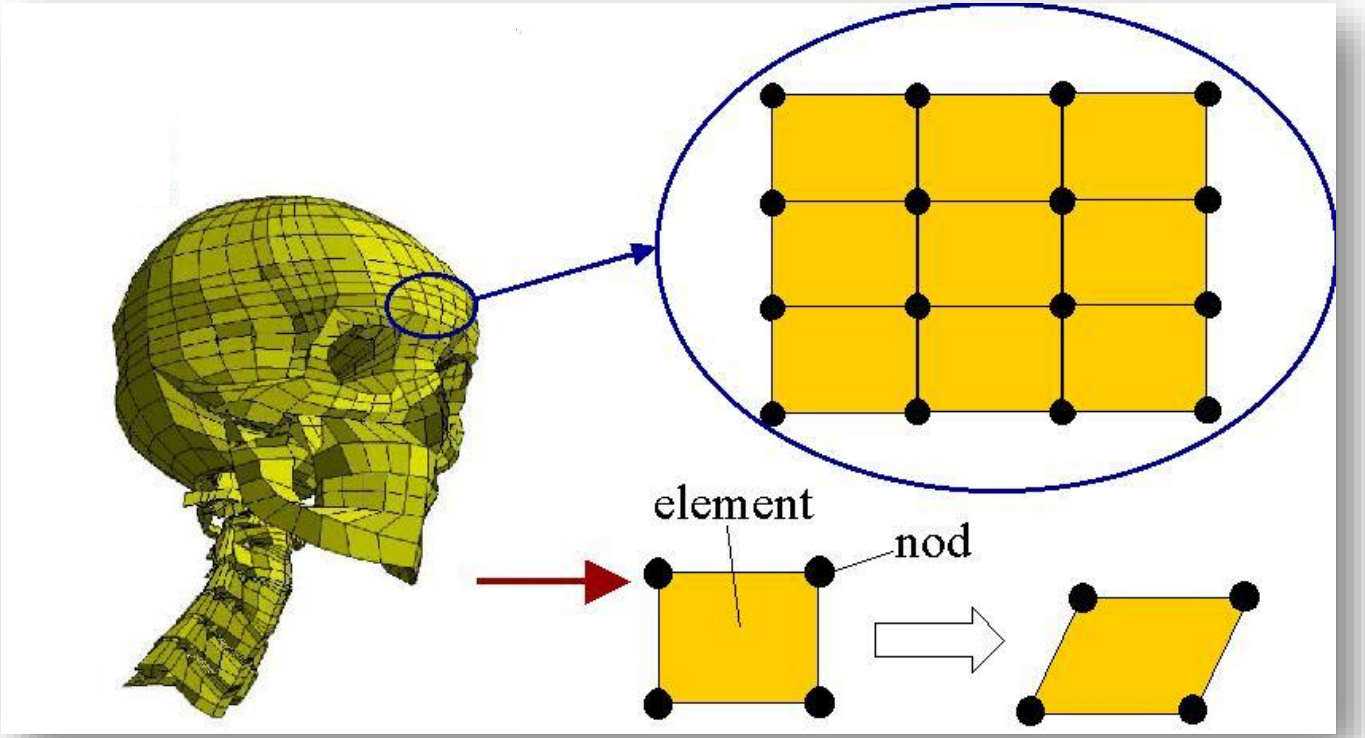
$$\mathbf{KU} = \mathbf{F}$$

Stiffness matrix | Nodal disp. | Nodal forces

You know all the equations, but you can not solve it by hand

Finite Element Analysis

1.1 FE mesh generation



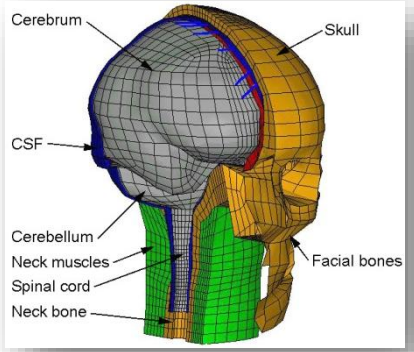
Finite Element Analysis

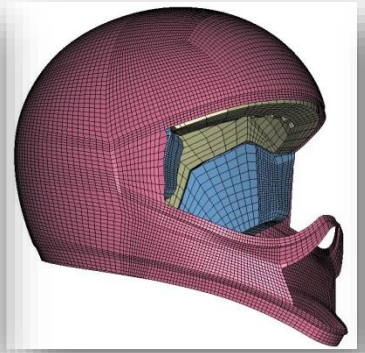
1.1 FE mesh generation

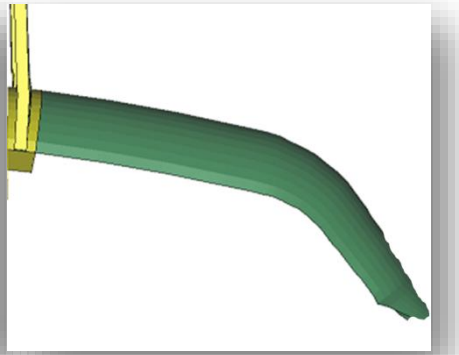
1.2 Material properties

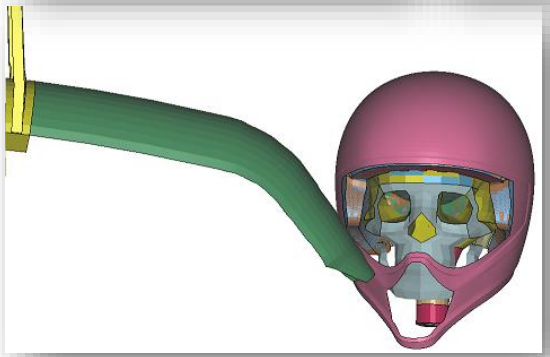
1.3 BC, load

2. Solve

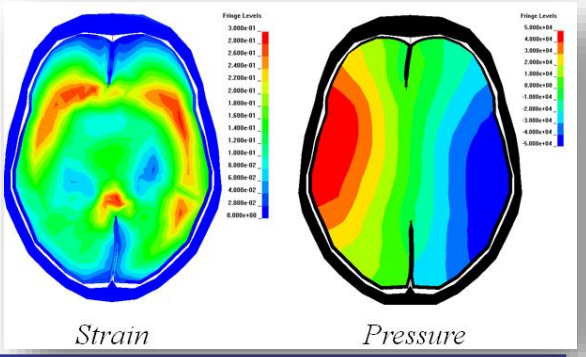








3. Analyze the results

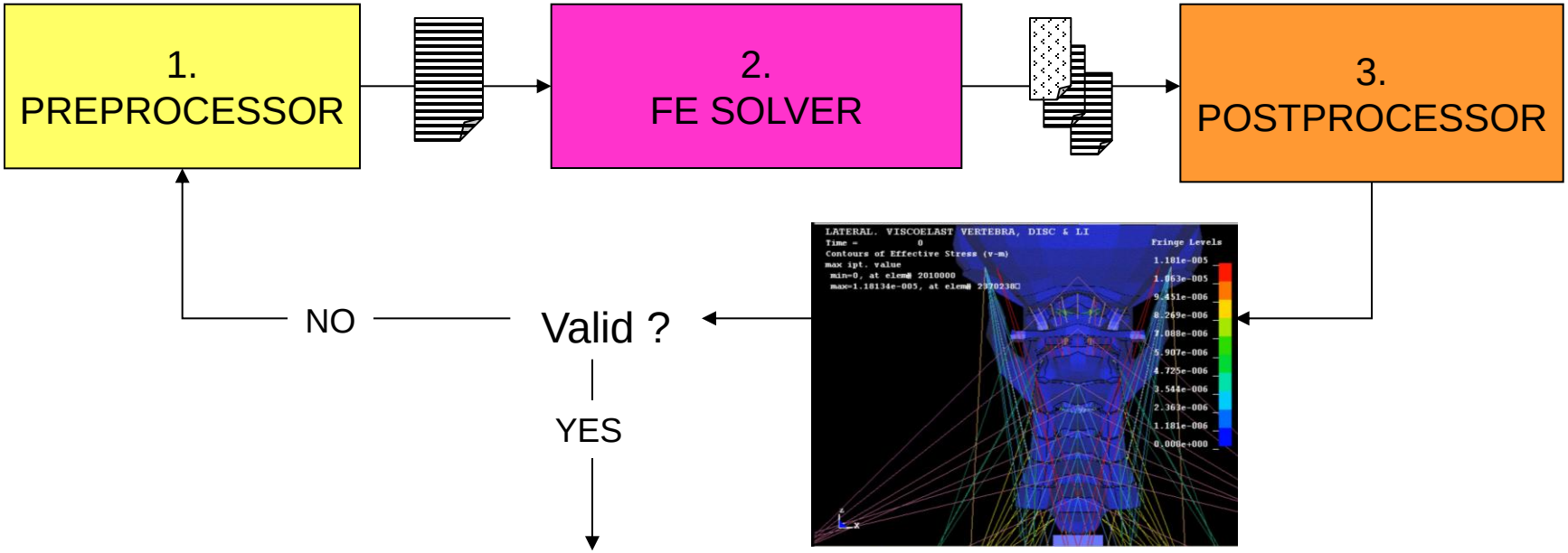


FEA Software

LS-DYNA, ABAQUS, ANSYS, COMSOL, ADINA...

Methodology

1. Describe problem (model, material, BC, load, equations…)
2. Solve using the finite element method
3. Analyze the results



"*All models are wrong, but some are still useful*."

George Box
Dept. Industrial Engineering
University of Wisconsin-Madison

How do we develop models that are useful?

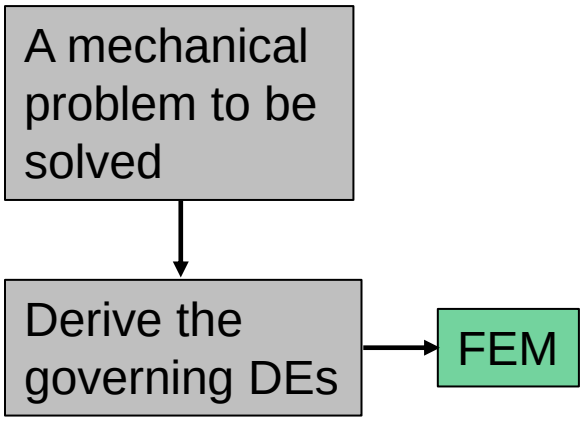
Verification & Validation

Verification = Are the mathematical equations correctly implemented?

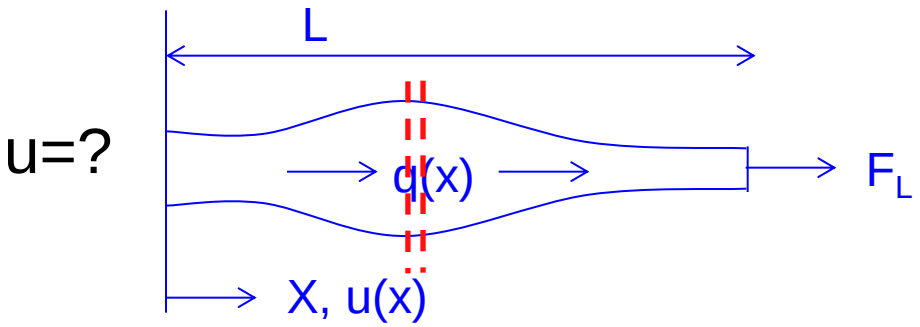
Validations = Does the FE model behave according to experimental data?

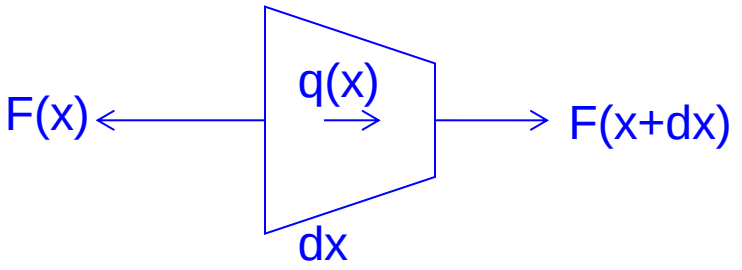
Linear Finite Elements

"How to perform a FE analysis from the early beginning to the very end"



Example, governing equations





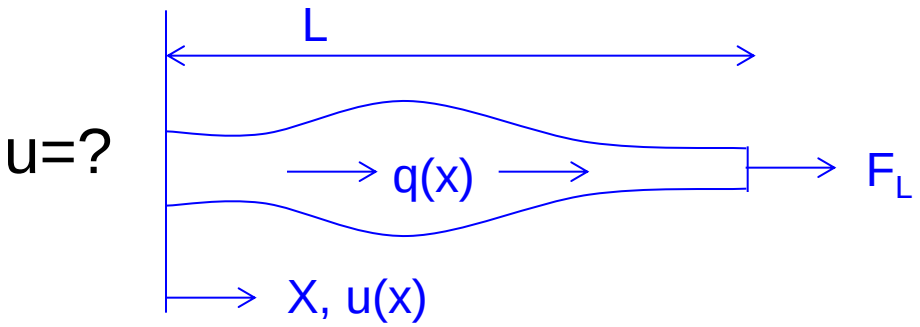
A(x): cross-section E(x): Young's modulus
q(x): body force

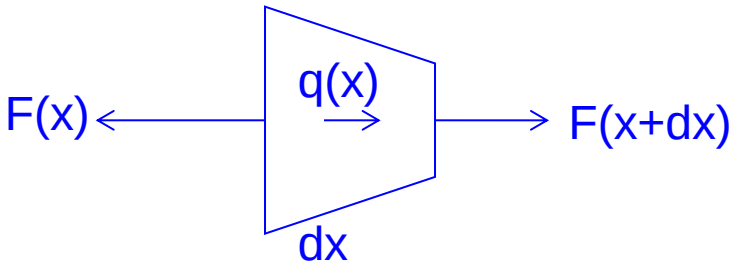
Equilibrium: $F(x) = q(x)dx + F(x+dx)$

$$-q(x) = \frac{F(x+dx) - F(x)}{dx}$$

$$\boxed{-q(x) = \frac{dF(x)}{dx}}$$

Example, governing equations





A(x): cross-section E(x): Young's modulus
q(x): body force

Equilibrium: $$-q(x) = \frac{dF(x)}{dx}$$

Stress: $$\sigma(x) = \frac{F(x)}{A(x)}$$

Constitutive Relation: $$\sigma(x) = E(x)\varepsilon(x)$$

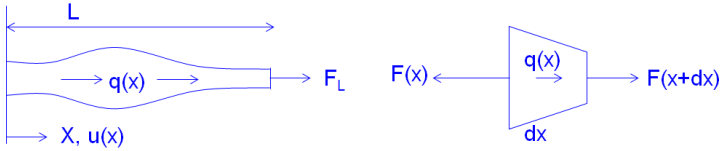
Strain: $$\varepsilon(x) = \frac{du(x)}{dx}$$

$$q(x) = -\frac{d}{dx}\left(E(x)A(x)\frac{du(x)}{dx}\right)$$

Boundary Conditions (BC): $u(0) = 0$, $F(L) = F_L$

Strong form
i.e. Original Differential Equation

Weak form



$$q(x) = -\frac{d}{dx}\left(E(x)A(x)\frac{du(x)}{dx} \right)$$

$v(x)$

Weight/test function

$$v(x)\left[\frac{d}{dx}\left(\mathrm{E}(x)A(x)\frac{du(x)}{dx} \right) + q(x) \right] = 0$$

Integrate

$$\int_0^L v(x)\left[\frac{d}{dx}\left(\mathrm{E}(x)A(x)\frac{du(x)}{dx} \right) + q(x) \right] dx = 0$$

$$\int u\,dv = uv - \int v\,du$$

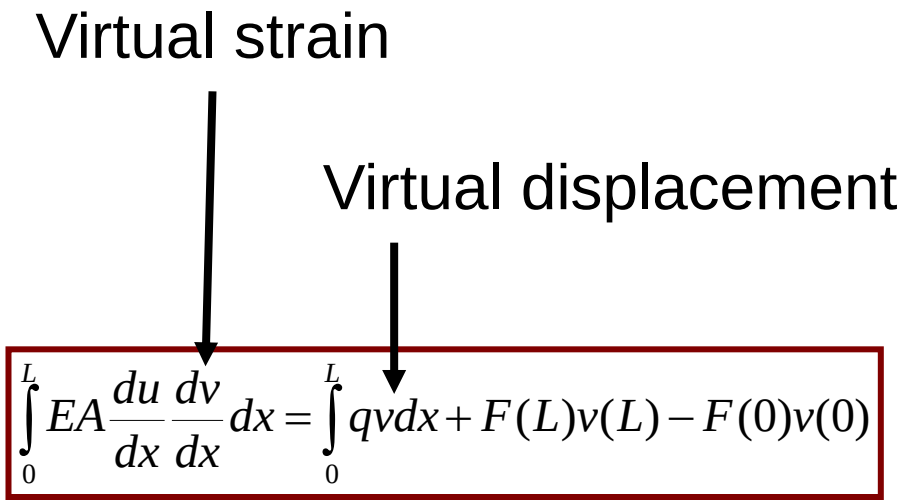
$$\left[\mathrm{E}\,A\frac{du}{dx}v \right]_0^L - \int_0^L EA\frac{du}{dx}\frac{dv}{dx}dx + \int_0^L qv dx = 0$$

$$\int_0^L EA\frac{du}{dx}\frac{dv}{dx}dx = \int_0^L qv dx + F(L)v(L) - F(0)v(0)$$

Weak form

Note: Weak form can be established

- *carrying out integration by parts from the strong form (PDE).*
- *directly according to the principle of virtual work*



Weak form

Note: Weak form can be established

- *carrying out integration by parts from the strong form (PDE).*
- *directly according to the principle of virtual work*

Internal virtual work

Virtual work due to boundary forces

$$\int_0^L EA\frac{du}{dx}\frac{dv}{dx}dx = \int_0^L qv dx + F(L)v(L) - F(0)v(0)$$

External virtual work

Weak form

(Virtual work form)

TASK: *In FEM the two concepts ”weak” and ”strong” formulations are used; what is meant with these two concepts?*

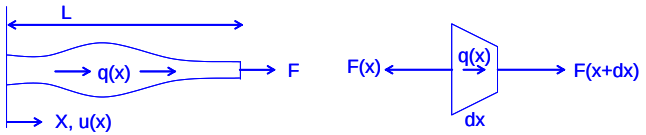
TASK: *In FEM the two concepts ”weak” and ”strong” formulations are used; what is meant with these two concepts?*

- Equlilibrium-> PDE : *strong form*
- Virtual work-> PDE of lower order+BC: *weak form*

Strong form: Displacement assumption must be at least twice diff., and satisfy the kinematic BC. -> u require *C^1 continuity. u' must be continuous and a second derivative of* u *must exist.*

Weak form: only a first derivative must exist and simpler functions may be used.
-> u and v require *C^0 continuity. u and v must be continuous; a first derivative must exist.*

Strong to weak form



$$q(x) = -\frac{d}{dx}\left(E(x)A(x)\frac{du(x)}{dx} \right)$$

BCs: $u(0) = 0$, $F(L) = F_L$

Strong form

$$\int_0^L EA\frac{du}{dx}\frac{dv}{dx}dx = \int_0^L qvdx + F(L)v(L) - F(0)v(0)$$

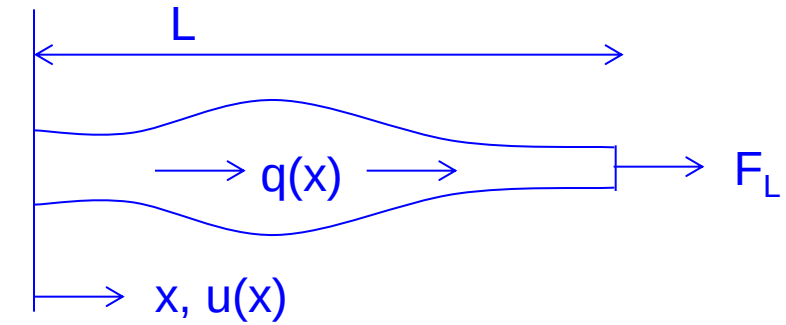
BCs: $u(0) = 0$

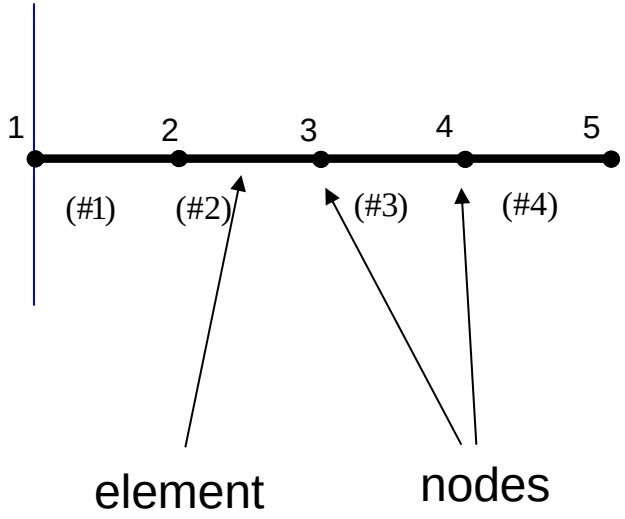
Weak form

To solve the problem with analytical methods we must find a displacement assumption that satisfies this equation and the BC!

Another strategy is to use FINITE ELEMENTS and find an **approximate** solution.

Finite elements





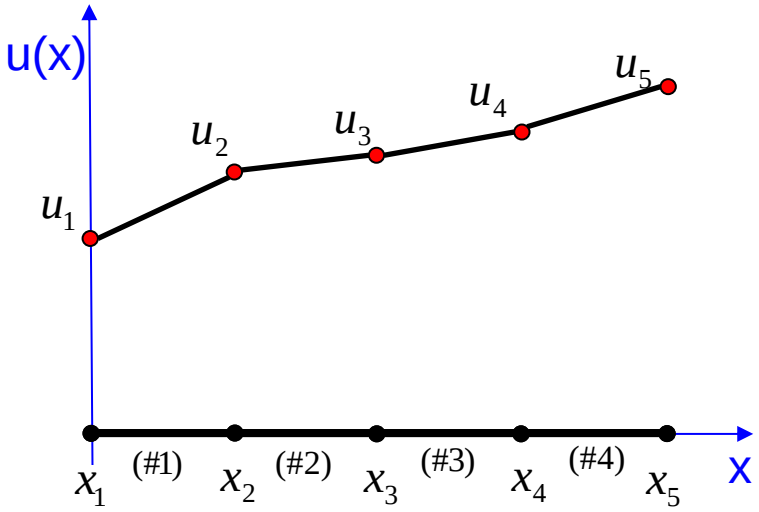
Simple assumptions for each element approximates the exact solution.

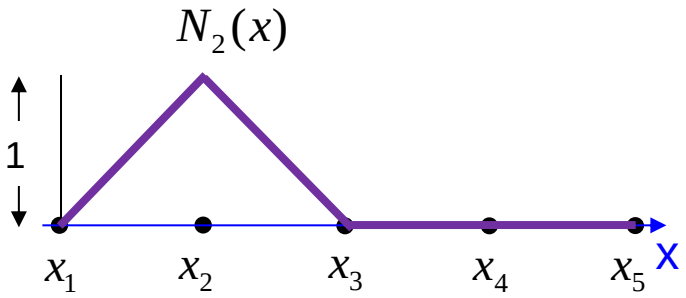
$$\int_0^L EA\frac{du}{dx}\frac{dv}{dx}dx = \int_0^L qvdx + F(L)v(L) - F(0)v(0) \quad \mathbf{KU = F}$$

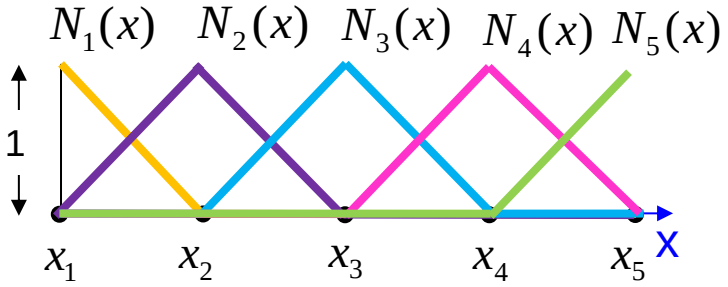


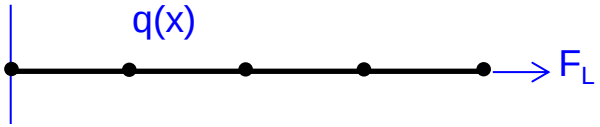
$$u(x) = u_1 N_1(x) + u_2 N_2(x) \ldots + u_5 N_5(x)$$

Trial fun. or shape fun. (1D linear element)









$$\int_0^L EA\frac{du}{dx}\frac{dv}{dx}dx = \int_0^L qvdx + F(L)v(L) - F(0)v(0) \quad \mathbf{KU = F}$$

$$u(x) = u_1N_1(x) + \ldots + u_5N_5(x)$$

$v(x)$ *(weight function/test)*

$$v_j(x) = N_j(x),\ j = 1,2,3,4,5$$

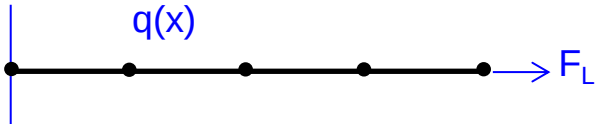
$$v_1(x) = N_1(x)$$

$$\int_0^L EA\frac{d(u_1N_1(x)+\ldots+u_5N_5(x))}{dx}\frac{dN_1(x)}{dx}dx = \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0)$$

$$\int_0^L EA\left(u_1\frac{dN_1(x)}{dx} + \ldots + u_5\frac{dN_5(x)}{dx}\right)\frac{dN_1(x)}{dx}dx = \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0)$$

$$u_1\int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_1(x)}{dx}dx + \ldots + u_5\int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_5(x)}{dx}dx = \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0)$$

$$\begin{bmatrix} \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_1(x)}{dx}dx & \ldots & \ldots & \ldots & \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_5(x)}{dx}dx \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0) \end{bmatrix}$$



$$u(x) = u_1 N_1(x) + \ldots + u_5 N_5(x)$$

$$\int_0^L EA \frac{du}{dx}\frac{dv}{dx}dx = \int_0^L qv dx + F(L)v(L) - F(0)v(0) \quad \mathbf{KU = F}$$

$$v_2(x) = N_2(x)$$

$$\int_0^L EA \frac{d(u_1 N_1(x) + \ldots + u_5 N_5(x))}{dx}\frac{dN_2(x)}{dx}dx = \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0)$$

$$\int_0^L EA\left(u_1\frac{dN_1(x)}{dx} + \ldots + u_5\frac{dN_5(x)}{dx}\right)\frac{dN_2(x)}{dx}dx = \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0)$$

$$u_1\int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_1(x)}{dx}dx + \ldots + u_5\int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_5(x)}{dx}dx = \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0)$$

$$\begin{bmatrix} \ldots & \ldots & \ldots & \ldots & \ldots \\ \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_1(x)}{dx}dx & \ldots & \ldots & \ldots & \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_5(x)}{dx}dx \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0) \\ \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0) \end{bmatrix}$$

$$\begin{bmatrix} \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_5(x)}{dx}dx \\ \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_5(x)}{dx}dx \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \int_0^L EA\frac{dN_5(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L EA\frac{dN_5(x)}{dx}\frac{dN_5(x)}{dx}dx \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0) \\ \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0) \\ \dots \\ \dots \\ \int_0^L qN_5(x)dx + F(L)N_5(L) - F(0)N_5(0) \end{bmatrix}$$

$$\begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} & K_{15} \\ K_{21} & K_{22} & K_{23} & K_{24} & K_{25} \\ K_{31} & K_{32} & K_{33} & K_{34} & K_{35} \\ K_{41} & K_{42} & K_{43} & K_{44} & K_{45} \\ K_{51} & K_{52} & K_{53} & K_{54} & K_{55} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{bmatrix}$$

$$\mathbf{KU} = \mathbf{F} \quad \Rightarrow \quad \mathbf{U} = \mathbf{K}^{-1}\mathbf{F}$$

Stiffness matrix

Nodal forces

Nodal displacements

E, A, L, h

$$\begin{bmatrix} \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_5(x)}{dx}dx \\ \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_5(x)}{dx}dx \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \int_0^L EA\frac{dN_5(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \end{bmatrix} = \begin{bmatrix} \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0) \\ \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0) \\ \dots \\ \dots \end{bmatrix}$$

q, F

$$E = 1,$$
$$A = 1,$$
$$L = 1,$$
$$F_L = 1,$$
$$q(x) = 1,$$
$$h = L/4$$

$$\begin{bmatrix} K_{11} \\ K_{21} \\ K_{31} \\ K_{41} & K_{42} & K_{43} & K_{44} & K_{45} \\ K_{5} \end{bmatrix} \begin{bmatrix} u_4 \end{bmatrix} = \begin{bmatrix} F_4 \end{bmatrix}$$

Give some numbers to make it more concrete!!

Skip during class… slides provided come to the final

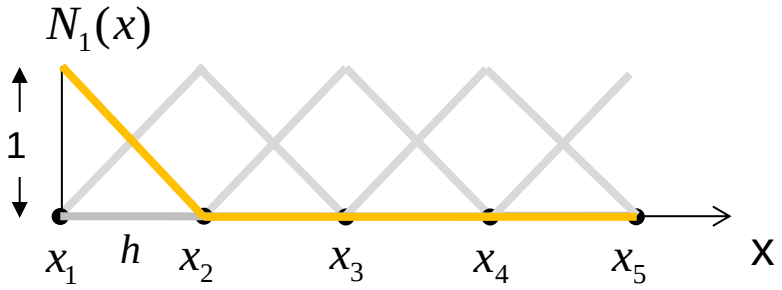
Stiffness matrix | Nodal forces

Nodal displacements

(assume E, A constant)

$$K_{11} = \int_0^L EA \frac{dN_1(x)}{dx} \frac{dN_1(x)}{dx} dx = \int_{x_1}^{x_2} EA \frac{dN_1(x)}{dx} \frac{dN_1(x)}{dx} dx$$

$$= \int_{x_1}^{x_2} EA \left(-\frac{1}{h}\right)\left(-\frac{1}{h}\right) dx = \frac{EA}{h}$$



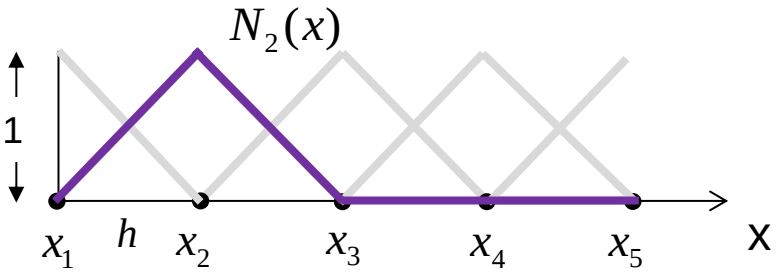
$$N_1(x) = \begin{cases} \dfrac{x_2 - x}{h} & x \in [x_1, x_2] \\ 0 & \text{otherwise} \end{cases}$$

$$K_{22} = \int_0^L EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx = \int_{x_1}^{x_3} EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx$$

$$= \int_{x_1}^{x_2} EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx + \int_{x_2}^{x_3} EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx$$

$$= 2\frac{EA}{h}$$

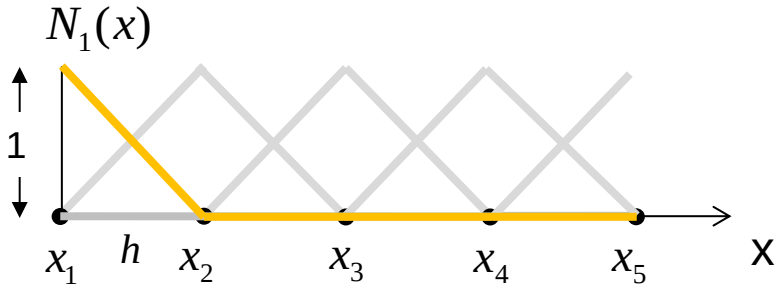
$$\begin{bmatrix} \frac{EA}{h} & & & & \\ & 2\frac{EA}{h} & & & \\ & & 2\frac{EA}{h} & & \\ & & & 2\frac{EA}{h} & \\ & & & & \frac{EA}{h} \end{bmatrix}$$



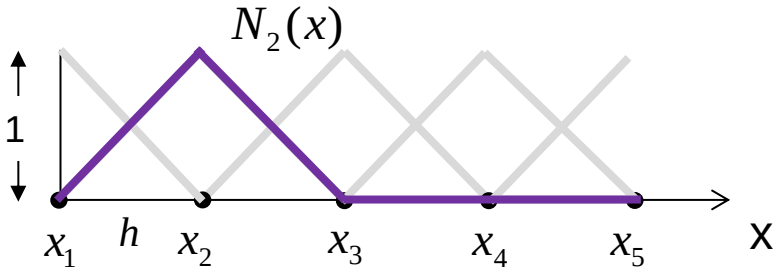
$$N_2(x) = \begin{cases} \dfrac{x - x_1}{h} & x \in [x_1, x_2] \\ \dfrac{x_3 - x}{h} & x \in [x_2, x_3] \\ 0 & \text{otherwise} \end{cases}$$

$$K_{21} = \int_0^L EA \frac{dN_2(x)}{dx} \frac{dN_1(x)}{dx} dx = \int_{x_1}^{x_2} EA \frac{dN_2(x)}{dx} \frac{dN_1(x)}{dx} dx$$

$$= \int_{x_1}^{x_2} EA \left(\frac{1}{h}\right)\left(-\frac{1}{h}\right) dx = -\frac{EA}{h}$$



$$N_1(x) = \begin{cases} \dfrac{x_2 - x}{h} & x \in [x_1, x_2] \\ 0 & \textit{otherwise} \end{cases}$$



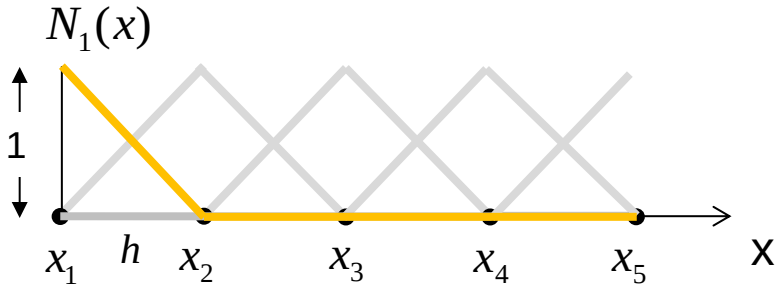
$$N_2(x) = \begin{cases} \dfrac{x - x_1}{h} & x \in [x_1, x_2] \\ \dfrac{x_3 - x}{h} & x \in [x_2, x_3] \\ 0 & \textit{otherwise} \end{cases}$$

$$K_{31} = \int_0^L EA \frac{dN_3(x)}{dx} \frac{dN_1(x)}{dx} dx = 0$$

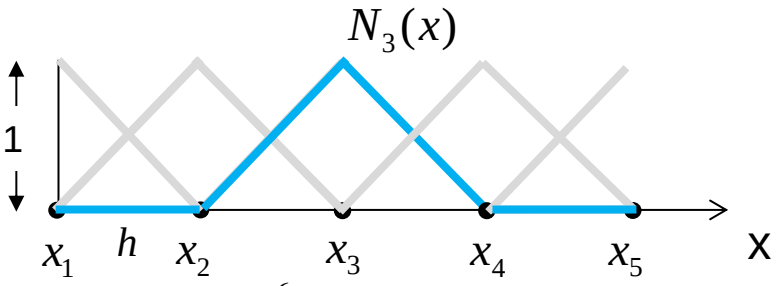
$$K_{41} = \int_0^L EA \frac{dN_4(x)}{dx} \frac{dN_1(x)}{dx} dx = 0$$

$$K_{51} = \int_0^L EA \frac{dN_5(x)}{dx} \frac{dN_1(x)}{dx} dx = 0$$

$$\begin{bmatrix} \frac{EA}{h} & -\frac{EA}{h} & 0 & 0 & 0 \\ -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} & 0 & 0 \\ 0 & -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} & 0 \\ 0 & 0 & -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} \\ 0 & 0 & 0 & -\frac{EA}{h} & \frac{EA}{h} \end{bmatrix}$$



$$N_1(x) = \begin{cases} \frac{x_2 - x}{h} & x \in [x_1, x_2] \\ 0 & otherwise \end{cases}$$



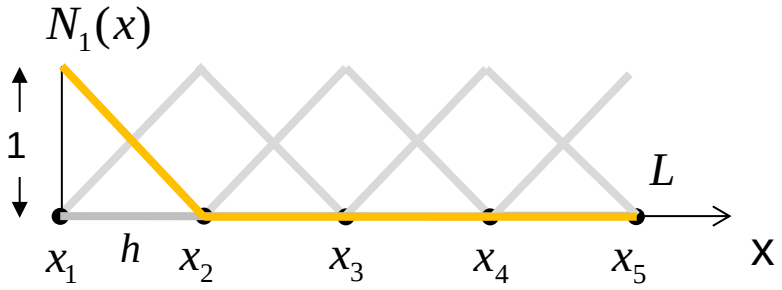
$$N_3(x) = \begin{cases} \frac{x - x_2}{h} & x \in [x_2, x_3] \\ \frac{x_4 - x}{h} & x \in [x_3, x_4] \\ 0 & otherwise \end{cases}$$

$$\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{bmatrix} = \begin{bmatrix} \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0) \\ \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0) \\ \\ \\ \int_0^L qN_5(x)dx + F(L)N_5(L) - F(0)N_5(0) \end{bmatrix}$$

$q = 1$

$$F_1 = \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0)$$

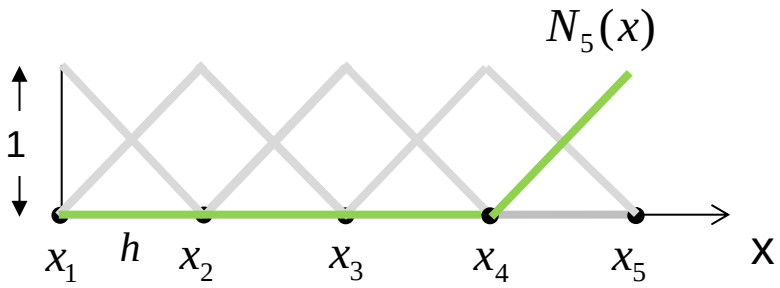
$$= \frac{h}{2} + F(L)*0 - F(0)*1 = \frac{h}{2} - F(0)$$



$$N_1(x) = \begin{cases} \frac{x_2 - x}{h} & x \in [x_1, x_2] \\ 0 & \textit{otherwise} \end{cases}$$

$$F_5 = \int_0^L qN_5(x)dx + F(L)N_5(L) - F(0)N_5(0)$$

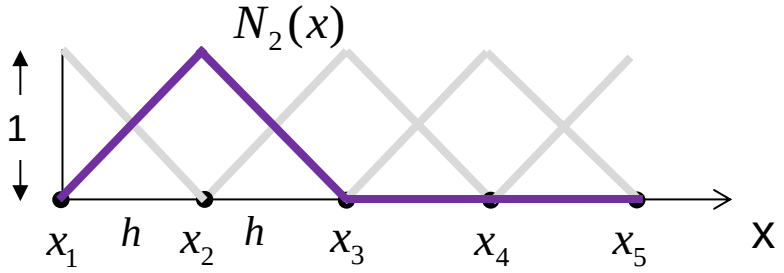
$$= \frac{h}{2} + F(L)*1 - F(0)*0 = \frac{h}{2} + F(L) = \frac{h}{2} + F_L$$



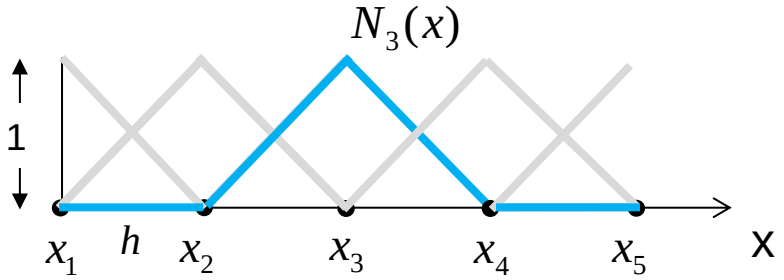
$q = 1$

$$F_2 = \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0)$$

$$= h + F(L)*0 - F(0)*0 = h$$



$$F_3 = F_4 = h$$



$$\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{bmatrix} = \begin{bmatrix} \frac{h}{2} - F(0) \\ h \\ h \\ h \\ \frac{h}{2} + F_L \end{bmatrix}$$

E, A, L, h

$$\begin{bmatrix} \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L EA\frac{dN_1(x)}{dx}\frac{dN_5(x)}{dx}dx \\ \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L EA\frac{dN_2(x)}{dx}\frac{dN_5(x)}{dx}dx \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \int_0^L EA\frac{dN_5(x)}{dx}\frac{dN_1(x)}{dx}dx & \dots & \dots & \dots & \int_0^L \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \end{bmatrix} = \begin{bmatrix} \int_0^L qN_1(x)dx + F(L)N_1(L) - F(0)N_1(0) \\ \int_0^L qN_2(x)dx + F(L)N_2(L) - F(0)N_2(0) \\ \dots \\ \dots \end{bmatrix}$$

q, F

$$E = 1,$$
$$A = 1,$$
$$L = 1,$$
$$F_L = 1,$$
$$q(x) = 1,$$
$$h = L/4$$

$$\begin{bmatrix} K_{11} \\ K_{21} \\ K_{31} \\ K_{41} & K_{42} & K_{43} & K_{44} & K_{45} \\ K_{5} \end{bmatrix} \begin{bmatrix} u_4 \end{bmatrix} = \begin{bmatrix} F_4 \end{bmatrix}$$

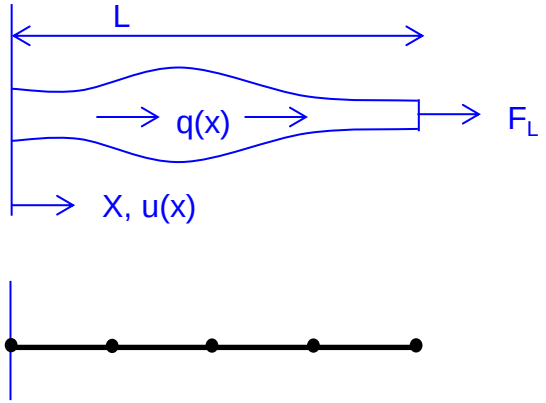
Give some numbers to make it more concrete!!

Skip during class… slides provided come to the final

Stiffness matrix | Nodal forces

Nodal displacements

$$\begin{bmatrix} \frac{EA}{h} & -\frac{EA}{h} & 0 & 0 & 0 \\ -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} & 0 & 0 \\ 0 & -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} & 0 \\ 0 & 0 & -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} \\ 0 & 0 & 0 & -\frac{EA}{h} & \frac{EA}{h} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} \frac{h}{2} - F(0) \\ h \\ h \\ h \\ \frac{h}{2} + F_L \end{bmatrix}$$



BCs: $u(0) = 0 \quad u_1 = 0$

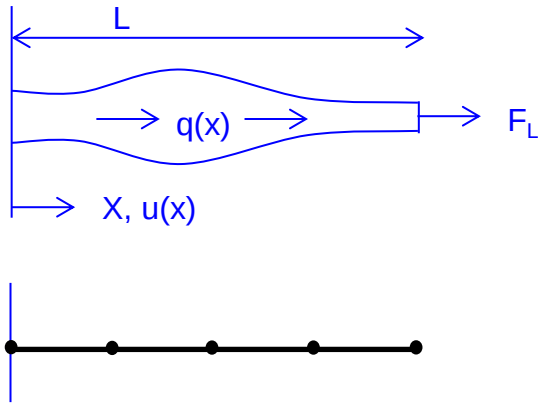
$F(L) = F_L$

$$\begin{bmatrix} 4 & -4 & 0 & 0 & 0 \\ -4 & 8 & -4 & 0 & 0 \\ 0 & -4 & 8 & -4 & 0 \\ 0 & 0 & -4 & 8 & -4 \\ 0 & 0 & 0 & -4 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0.125 - F(0) \\ 0.25 \\ 0.25 \\ 0.25 \\ 1.125 \end{bmatrix}$$

$E = 1, A = 1, L = 1, F_L = 1, q(x) = 1$

$h = L / 4$

$$\begin{bmatrix} \frac{EA}{h} & -\frac{EA}{h} & 0 & 0 & 0 \\ -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} & 0 & 0 \\ 0 & -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} & 0 \\ 0 & 0 & -\frac{EA}{h} & 2\frac{EA}{h} & -\frac{EA}{h} \\ 0 & 0 & 0 & -\frac{EA}{h} & \frac{EA}{h} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} \frac{h}{2} - F(0) \\ h \\ h \\ h \\ \frac{h}{2} + F_L \end{bmatrix}$$



BCs: $u(0) = 0 \quad u_1 = 0$

$F(L) = F_L$

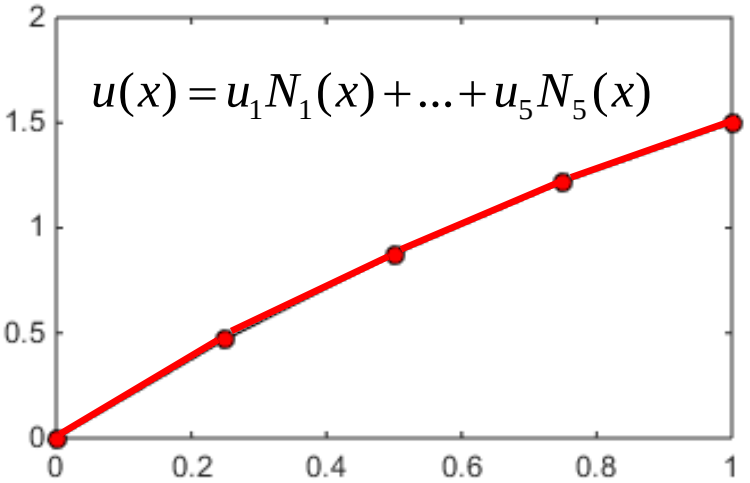
$$\begin{bmatrix} 4 & -4 & 0 & 0 & 0 \\ -4 & 8 & -4 & 0 & 0 \\ 0 & -4 & 8 & -4 & 0 \\ 0 & 0 & -4 & 8 & -4 \\ 0 & 0 & 0 & -4 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0.125 - F(0) \\ 0.25 - (-4)u_1 \\ 0.25 - (0)u_1 \\ 0.25 - (0)u_1 \\ 1.125 - (0)u_1 \end{bmatrix}$$

$E = 1, A = 1, L = 1, F_L = 1, q(x) = 1$

$h = L / 4$

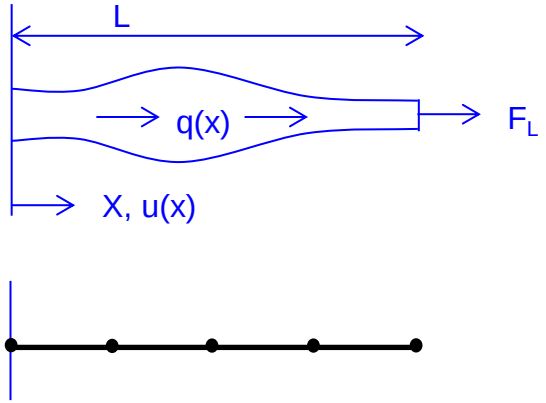
Stiffness matrix elimination for B.C. ($u_1 = 0$)

$$\begin{bmatrix} 8 & -4 & 0 & 0 \\ -4 & 8 & -4 & 0 \\ 0 & -4 & 8 & -4 \\ 0 & 0 & -4 & 4 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 1.125 \end{bmatrix} \quad \begin{bmatrix} u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0.46875 \\ 0.875 \\ 1.21875 \\ 1.50 \end{bmatrix}$$



Gaussian Elimination or Matlab

$$F(0) = ?$$



$$4u_1 - 4u_2 = 0.125 - F(0)$$
$$F(0) = 2$$

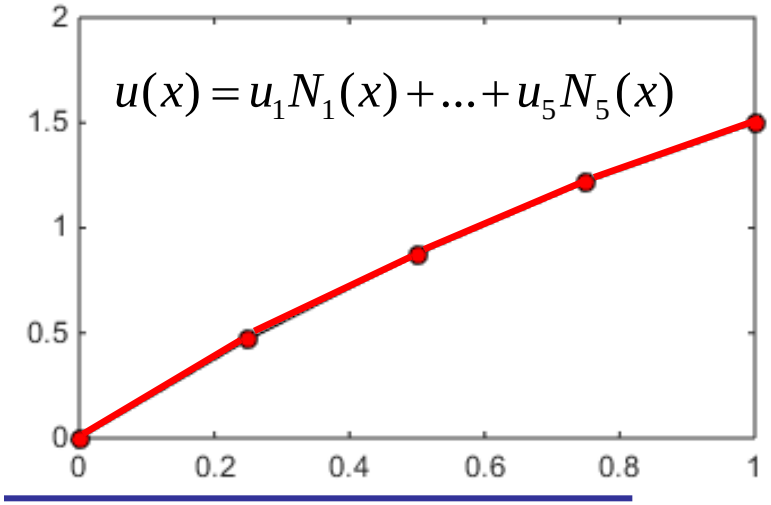
BCs: $u(0) = 0 \quad u_1 = 0$
$F(L) = F_L$

$$\begin{bmatrix} 4 & -4 & 0 & 0 & 0 \\ -4 & 8 & -4 & 0 & 0 \\ 0 & -4 & 8 & -4 & 0 \\ 0 & 0 & -4 & 8 & -4 \\ 0 & 0 & 0 & -4 & 4 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0.125 - F(0) \\ 0.25 - (-4)u_1 \\ 0.25 - (0)u_1 \\ 0.25 - (0)u_1 \\ 1.125 - (0)u_1 \end{bmatrix}$$

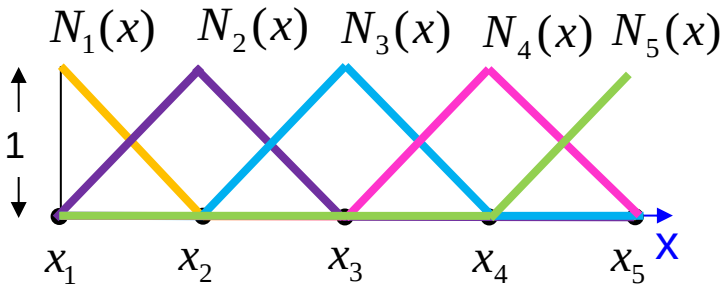
$E = 1, A = 1, L = 1, F_L = 1, q(x) = 1$
$h = L / 4$

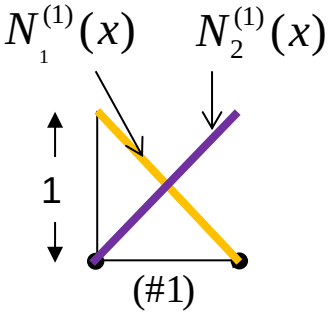
Stiffness matrix elimination for B.C. ($u_1 = 0$)

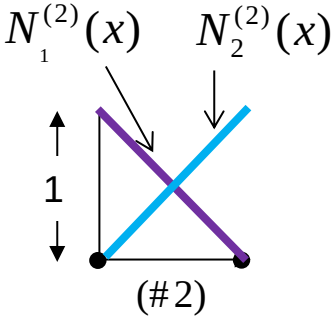
$$\begin{bmatrix} 8 & -4 & 0 & 0 \\ -4 & 8 & -4 & 0 \\ 0 & -4 & 8 & -4 \\ 0 & 0 & -4 & 4 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 1.125 \end{bmatrix} \quad \begin{bmatrix} u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} 0.46875 \\ 0.875 \\ 1.21875 \\ 1.50 \end{bmatrix}$$



Gaussian Elimination or Matlab







$$K_{11} = \int_0^L EA \frac{dN_1(x)}{dx} \frac{dN_1(x)}{dx} dx$$

$$K_{11}^{(1)} = \int^{(\#1)} EA \frac{dN_1^{(1)}(x)}{dx} \frac{dN_1^{(1)}(x)}{dx} dx$$

$$K_{22} = \int_0^L EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx = \int_{x_1}^{x_3} EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx$$

$$= \int_{x_1}^{x_2} EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx + \int_{x_2}^{x_3} EA \frac{dN_2(x)}{dx} \frac{dN_2(x)}{dx} dx$$

$$K_{22} = \int^{(\#1)} EA \frac{dN_2^{(1)}(x)}{dx} \frac{dN_2^{(1)}(x)}{dx} dx + \int^{(\#2)} EA \frac{dN_1^{(2)}(x)}{dx} \frac{dN_1^{(2)}(x)}{dx} dx$$

$$= K_{22}^{(1)} + K_{11}^{(2)}$$

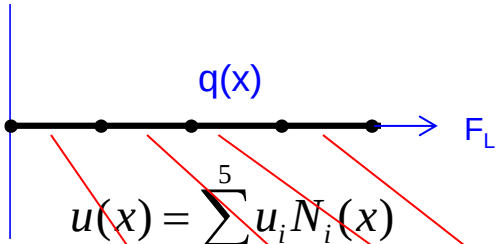
$$K = \begin{bmatrix} K_{11} & K_{12} & 0 & 0 & 0 \\ K_{21} & K_{22} & K_{23} & 0 & 0 \\ 0 & K_{32} & K_{33} & K_{34} & 0 \\ 0 & 0 & K_{43} & K_{44} & K_{45} \\ 0 & 0 & 0 & K_{54} & K_{55} \end{bmatrix}$$

Global

$$\begin{bmatrix} K_{11}^{(1)} & K_{12}^{(1)} & & & \\ K_{21}^{(1)} & K_{22}^{(1)} + K_{11}^{(2)} & K_{12}^{(2)} & & \\ & K_{21}^{(2)} & K_{22}^{(2)} + K_{11}^{(3)} & K_{12}^{(3)} & \\ & & K_{21}^{(3)} & K_{22}^{(4)} + K_{11}^{(4)} & K_{12}^{(4)} \\ & & & K_{21}^{(4)} & K_{22}^{(4)} \end{bmatrix}$$

Local

Stiffness matrix assembly



$$\mathbf{KU} = \mathbf{F}$$

$$\begin{bmatrix} K_{11}^{(1)} & K_{12}^{(1)} & 0 & 0 & 0 \\ K_{21}^{(1)} & K_{22}^{(1)} + K_{11}^{(2)} & K_{12}^{(2)} & 0 & 0 \\ 0 & K_{21}^{(2)} & K_{22}^{(2)} + K_{11}^{(3)} & K_{12}^{(3)} & 0 \\ 0 & 0 & K_{21}^{(3)} & K_{22}^{(4)} + K_{11}^{(4)} & K_{12}^{(4)} \\ 0 & 0 & 0 & K_{21}^{(4)} & K_{22}^{(4)} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \\ F_5 \end{bmatrix}$$

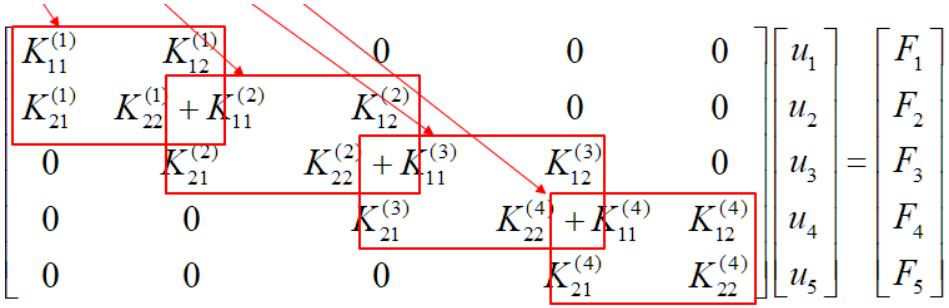
Development of Finite Element Equations

Matrix form: $\mathbf{KU} = \mathbf{F} \Rightarrow \mathbf{U} = \mathbf{K}^{-1}\mathbf{F}$

Direct Stiffness Method (Computer Lab FEM)

Based on physical reasoning (force equilibrium) and limited to simple cases (e.g. 1D bar or beam analysis)

Element stiffness matrix need to be converted to global coordinate system before assembly because **U, F** are in global.

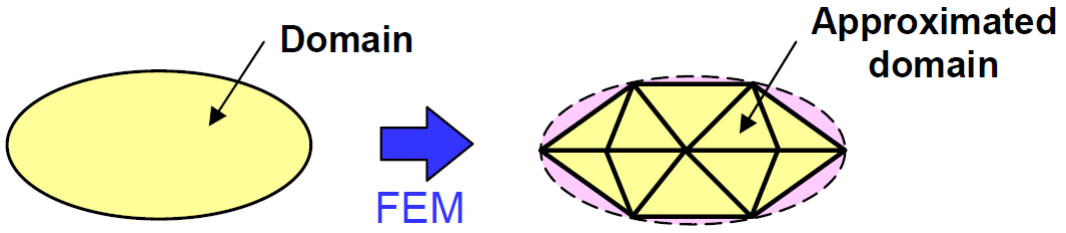


Useful literature for Lab B

1. Bathe, K-J, Finite Element Procedures, Upper Saddle River USA 1996 or later, Prentice-Hall Inc. Chapter 3.2.1 (Direct Stiffness Method)

Potential Sources of Error in FEA

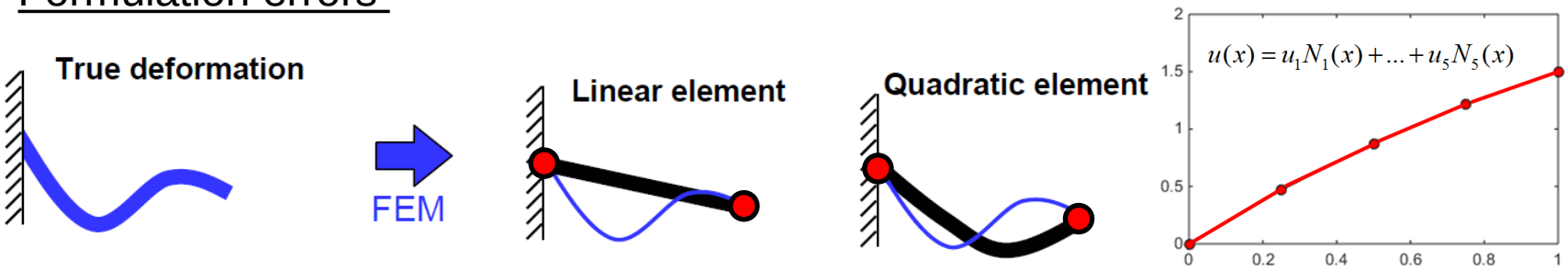
- Geometry is simplified



Discretization errors

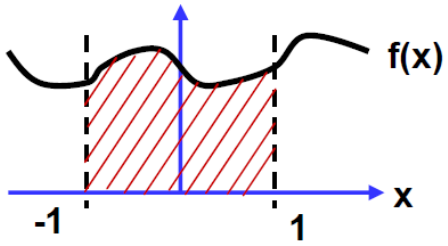
- Field quantify is assumed to be polynomial over an element (may not be true)

Formulation errors



- Use numerical integration techniques (e.g. Gauss integration)

$$K_{11} = \int_0^L EA \frac{dN_1(x)}{dx} \frac{dN_1(x)}{dx} dx$$



- Round-off error: computer carries only a finite number of digits

Generalization

- 1D –> 3D
- Static –> Dynamic
- In practice ...

1D linear shape function

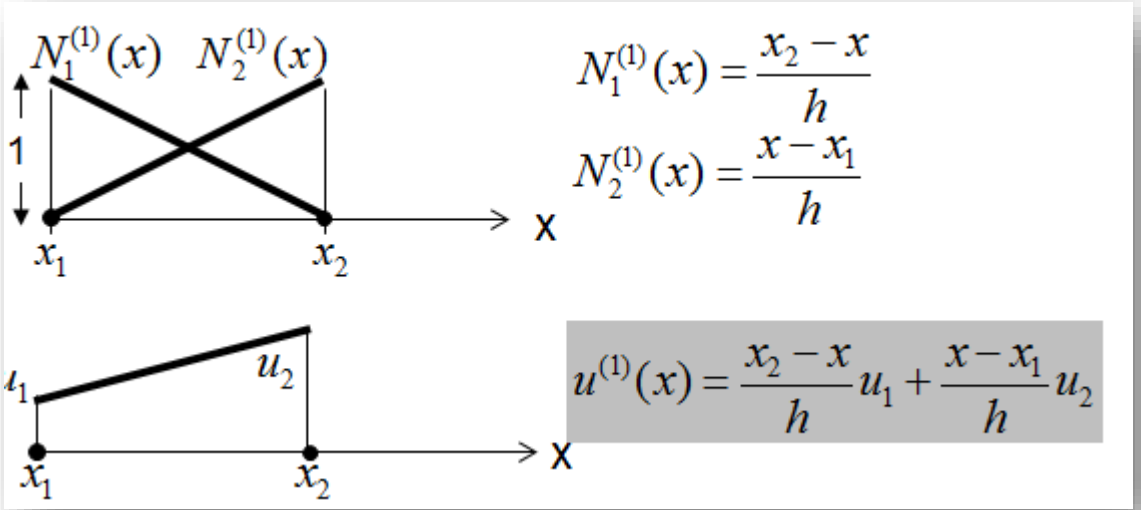
Assume piecewise linear in an element

$$u^{(1)}(x) = \sum_{i=1}^{2} N_i^{(1)}(x) u_i$$

$$u^{(1)}(x) = N_1^{(1)}(x) u_1 + N_2^{(1)}(x) u_2$$

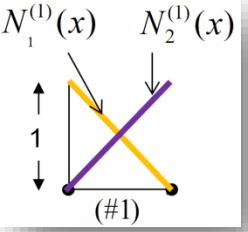
$$u^{(1)}(x) = [N_1^{(1)}(x)\ N_2^{(1)}] \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Element #1

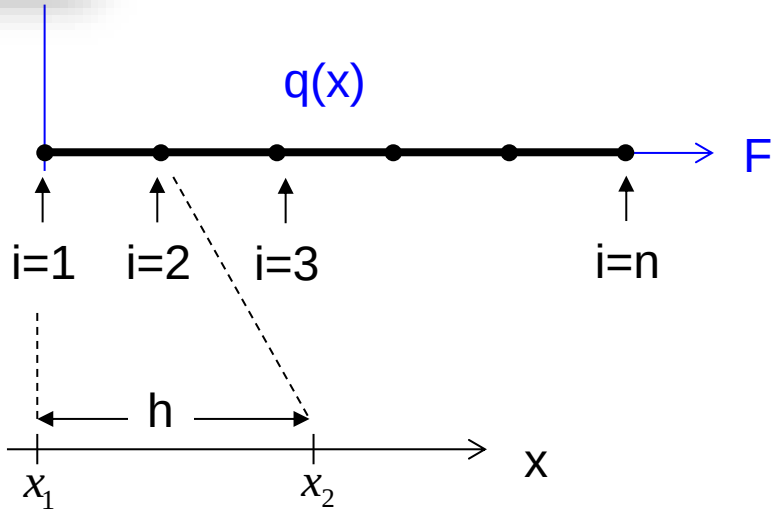


$$u^{(1)}(x) = \mathbf{N}^{(1)}(x)\mathbf{U}$$ Elem#1

$$u^{(m)}(x) = \mathbf{N}^{(m)}(x)\mathbf{U}$$ Elem#m



1D linear shape function

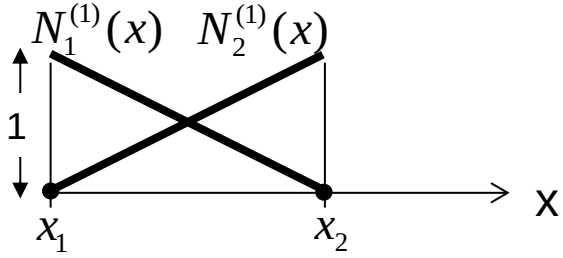


Assume piecewise linear in an element

$$u^{(1)}(x) = \sum_{i=1}^{2} N_i^{(1)}(x) u_i$$

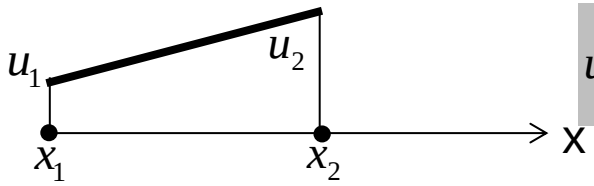
u(x) at the nodes

Element shape function



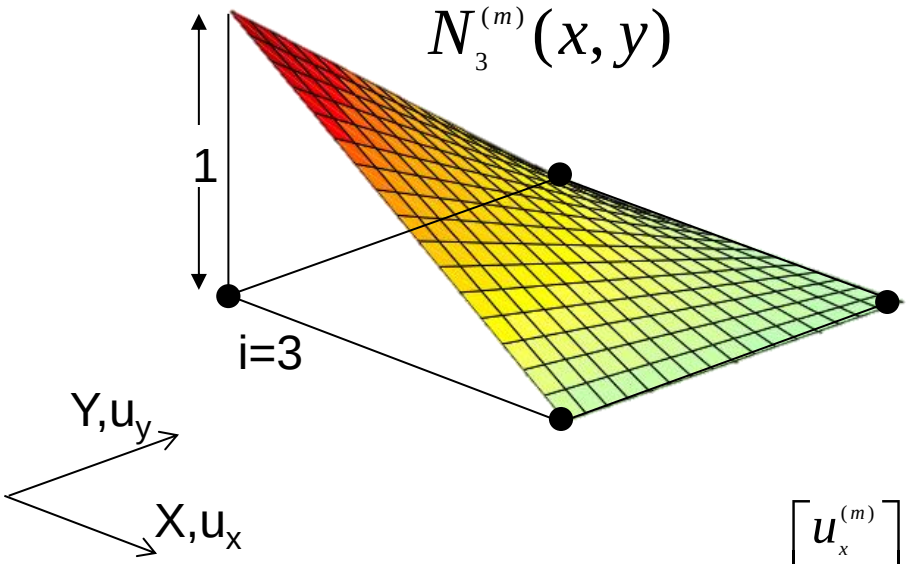
$$N_1^{(1)}(x) = \frac{x_2 - x}{h}$$

$$N_2^{(1)}(x) = \frac{x - x_1}{h}$$



$$u^{(1)}(x) = \frac{x_2 - x}{h} u_1 + \frac{x - x_1}{h} u_2$$

2D linear shape function



Assume piecewise linear in an element

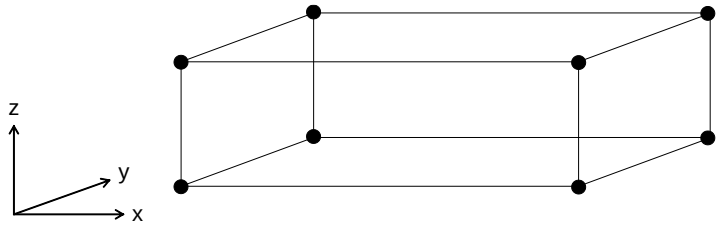
$$u_x^{(m)}(x,y) = \sum_{i=1}^{4} N_i^{(m)}(x,y) u_{ix}$$

$$u_y^{(m)}(x,y) = \sum_{i=1}^{4} N_i^{(m)}(x,y) u_{iy}$$

$$\begin{bmatrix} u_x^{(m)} \\ u_y^{(m)} \end{bmatrix} = \begin{bmatrix} N_1^{(m)} & 0 & N_2^{(m)} & 0 & N_3^{(m)} & 0 & N_4^{(m)} & 0 \\ 0 & N_1^{(m)} & 0 & N_2^{(m)} & 0 & N_3^{(m)} & 0 & N_4^{(m)} \end{bmatrix} \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \end{bmatrix}$$

$$\mathbf{u}(x,y) = \mathbf{N}^{(m)}(x,y)\mathbf{U}$$

$$\mathbf{u} = \mathbf{N}^{(m)}\mathbf{U}$$



Displacements

$$u^{(m)}(x,y,z) = N^{(m)}(x,y,z)U$$

Some of these are unknown and some are known due to BC.

$$U = \begin{bmatrix} u_{1x} \\ u_{1y} \\ \\ \\ u_{n\,xrot} \\ \\ \\ \end{bmatrix}$$

References

[1] Dynamic Finite Element Methods. Lecture notes for Biomechanics and Neuronics, By Svein Kleiven, Peter Halldin and Dan Zenkert (Bilda)

[2] *Bathe, K-J, Finite Element Procedures, Upper Saddle River USA 1996 or later, Prentice-Hall Inc*. Chapter 3.2.1 (Direct Stiffness Method)

[3] Gilbert Strang: *on the history of the finite element method (full 11 min)*
https://www.youtube.com/watch?v=WwgrAH-IMOk

[4] Gilbert Strang: Finite Elements in 1D

Lecture 17 (from 37:00) (part 1), Lecture 18 (part 2)
https://ocw.mit.edu/courses/mathematics/18-085-computational-science-and-engineering-i-fall-2008/video-lectures/